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Review article

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Bidirectional brain–heart interactions in health and disease

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Abstract

The brain and heart constantly communicate with each other to adapt to changes in the environment. In recent years, we have seen increasing recognition that brain–heart interactions influence many cognitive and behavioural functions, and that disturbances in this bidirectional relationship can have detrimental effects on both organs. Clinically, such disturbances can manifest as cardiac dysfunction following acute psychological stress or a brain injury such as acute ischaemic stroke, or as cognitive disorders in patients with acute or chronic heart disease. In this Review, we provide an overview of the physiological interaction between the heart and the brain and show how injury or dysfunction in one of these organs can lead to dysfunction in the other. We describe the current state of knowledge on neuronal correlates and humoral mediators of brain–heart interactions, as well as state-of-the-art methods for studying these interactions. We highlight the clinical implications of disturbances in physiological brain–heart communication for various clinical conditions, including Takotsubo syndrome resulting from emotional stress or from brain disorders, cardiac complications after stroke (so-called stroke–heart syndrome), and cognitive dysfunction in patients with cardiac diseases such as heart failure and atrial fibrillation. In addition, we address sex and gender differences in this bidirectional relationship.

Sections

Introduction

Physiological brain–heart interactions

Emotional stress and cardiovascular disease

Brain–heart interactions in stroke

Heart disease and cognitive impairment

Conclusions and implications for patient care

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Key points

- Heart and brain function are closely interconnected through neural, humoral and mechanical brain–heart circuits.
- Pathological interactions between the brain and the heart have key pathophysiological roles in myocardial ischaemia and dysfunction in patients with Takotsubo syndrome.
- Acute ischaemic brain injury disrupts physiological brain–heart signalling through the autonomic nervous system, the hypothalamic–pituitary–adrenal axis and the immune system, potentially leading to heart damage and dysfunction (the so-called stroke–heart syndrome).
- Cardiac diseases such as heart failure, atrial fibrillation and myocardial infarction impair neurohormonal and neuromechanical communication between the heart and the brain, which can lead to cognitive impairment and dementia.
- Sex and gender differences in stress circuits, autonomic regulation and immune responses contribute to distinct patterns of neurological and cardiovascular diseases.

Introduction

The global burden of brain and heart diseases is a considerable public health challenge, with both conditions contributing substantially to morbidity, disability and mortality worldwide[1,2]. As the prevalence of these diseases continues to rise globally[3], understanding the mechanisms that underlie their interplay is increasingly important. The heart and brain interact in the physiological regulation of cardiovascular parameters and are mutually involved in various other physiological processes, including the regulation of cognitive functions and emotion[4,5]. Importantly, brain–heart interactions have a crucial role in numerous disease states, such as following acute brain injuries such as stroke, which can lead to heart damage[6,7]. Conversely, heart disease can have complex effects on brain health, contributing to cognitive impairments, psychiatric disorders and other neuropsychiatric sequelae[8,9].

This Review explores how reciprocal interactions between the heart and brain can cause dysfunction in one of these organs to adversely affect the other. We begin by describing the physiological communication between the heart and the brain, and the methods that are used to measure these interactions. Drawing on current knowledge, including neuronal correlates and humoral mediators, we shed light on the clinical implications of disrupted brain–heart communication. We discuss clinical conditions, such as Takotsubo syndrome (TTS) and stroke–heart syndrome (SHS), in which these interactions are particularly prominent. In addition, we address the critical roles of sex and gender differences in brain–heart interactions, highlighting the importance of a comprehensive understanding of the brain–heart axis for improving patient outcomes. Of note, mechanisms that lead to direct interorgan damage, such as strokes caused by atrial fibrillation or cardiac arrest leading to hypoxic brain injury, are beyond the scope of this Review.

Physiological brain–heart interactions

The brain and heart communicate continuously through multiple pathways that operate on different time scales. The core constituents are control loops that maintain cardiovascular parameters within certain physiological limits with the aim of providing adequate blood flow to the brain and body while adapting to changing demands. Afferent input from stretch-sensitive (and, hence, pressure-indicating) baroreceptors in large arteries ('high-pressure' baroreceptors), atria and venous vessels ('low-pressure' baroreceptors) are relayed to the brainstem through the vagal and glossopharyngeal nerves[10].

An important relay station is the nucleus tractus solitarius, which receives input from high-pressure baroreceptors, leading to increased parasympathetic activity and decreased sympathetic activity[10]. This control loop (Fig. 1) allows heart rate, heart contractility, blood pressure and other cardiovascular parameters to be adjusted within seconds to meet changing needs, such as standing up, lying down or exercising. It also enables temporary adjustments of the target values ('setpoints') for cardiovascular parameters; for example, a temporary increase in blood pressure can be achieved during stress by transiently reducing baroreceptor sensitivity[11]. Long-term adjustments of setpoints occur during the development of arterial hypertension[12].

A second major communication and cardiovascular control system operates through the secretion of hormones and other bioactive molecules into the bloodstream. For example, during stress, the hypothalamus increases sympathetic outflow to the adrenal glands, which release adrenaline, noradrenaline and cortisol into the bloodstream to act on the heart and the cardiovascular system. Conversely, cardiomyocytes release natriuretic peptides, which counteract the sympathetic nervous system (SNS), modulate the renin–angiotensin–aldosterone system and affect brain integrity through receptors in the CNS[13,14]. Other examples of humoral communication between the brain and heart include neuronal release of neuropeptide Y, which modulates the stress response[15], and release of extracellular vesicles (EVs) — which transport bioactive molecules — from cardiac or CNS cells[7,16].

Additional mechanical pathways of brain–heart communication have also been identified. For example, a study published in 2024 indicated that heartbeat-related effects on neural activity are mediated directly by interaction of the pulse with central neurons through mechanosensitive ion channels (most probably PIEZO2)[17]. The interplay of neural, humoral and mechanical pathways of the brain–heart axis have been extensively reviewed elsewhere[18].

The hypothalamus and brainstem nuclei are key components of the control circuits of the brain–heart axis and are structurally and functionally connected to higher brain areas. Together, they constitute the central autonomic network (CAN), which comprises the following brain structures: nucleus tractus solitarius; ventrolateral medulla; parabrachial complex (dorsal pons); periaqueductal grey; hypothalamus (lateral hypothalamic nuclei); dorsal anterior cingulate cortex; genual and subgenual cingulate cortices; anterior, mid and posterior insular cortices; orbitofrontal cortex; and amygdala[19,20] (Fig. 1). Importantly, areas of the CAN are known to be involved in processing of emotions, and also other cognitive functions such as conflict processing and multisensory awareness[20]. The close anatomical connections between these central brain areas and the visceral control units of the heart (and the rest of the body) indicate an intrinsic connection between the emotional and cognitive functions of these brain areas and the heart.

Effects of emotional and cognitive function on cardiac function, and vice versa, can be observed on short time scales (within seconds). For example, transcranial magnetic stimulation (TMS) of the dorsolateral prefrontal cortex can induce rapid deceleration of heart rate[21]. A physiological example of such a top-down influence on the heart is the well-established orientation response, which occurs when a

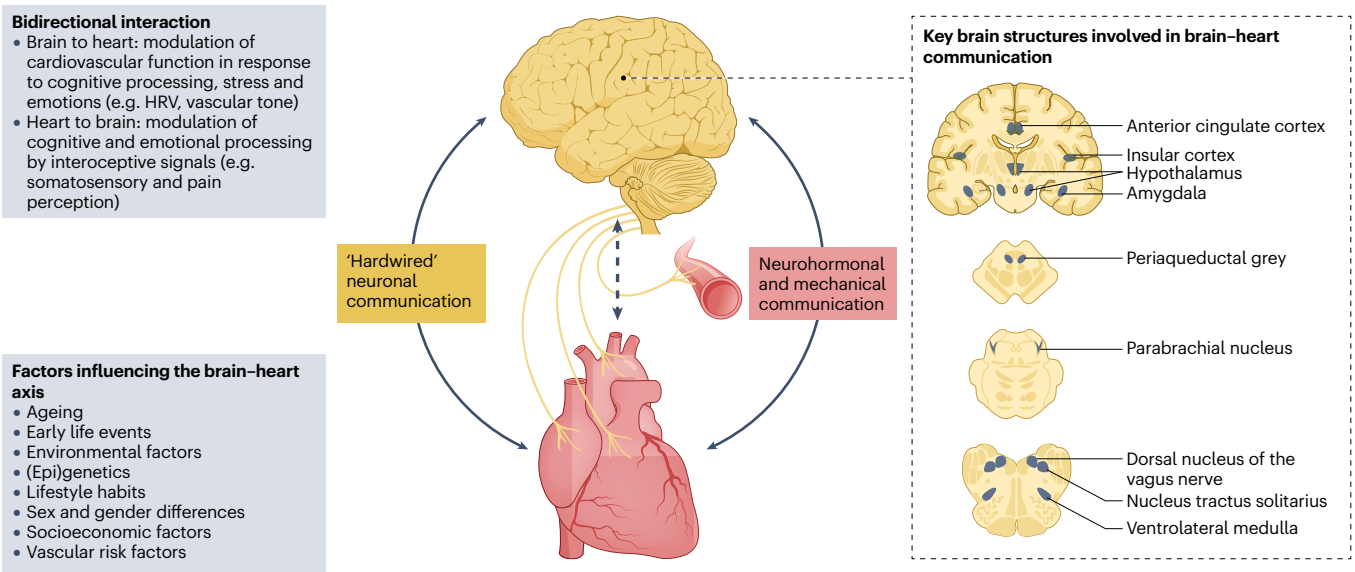


Fig. 1 | Physiological bidirectional brain–heart communication. The heart and brain communicate through neuronal, neurohormonal and mechanical pathways, resulting in bidirectional modulation of cardiovascular and cognitive functions. HRV, heart rate variability.

person is exposed to novel stimuli and also consists of transient heart rate deceleration[22]. As examples of bottom-up effects, structured modulation of somatosensory perception[5], pain perception[23], motor excitability[24], emotional processing[25,26] and many other cognitive and emotional functions can be observed during each cardiac cycle in the period between increased baroreceptor activity (systole) and low baroreceptor activity (diastole). The mediating role of baroreceptors in large arterial vessels is supported by the finding that direct baroreceptor stimulation can alter pain perception[27].

Owing to their rapid and continuous bidirectional interactions and the fact that they belong to tightly interconnected control circuits, the activities of the brain and heart are so tightly coupled in most natural situations that unidirectional causality cannot be assumed[9]. In the field of emotion research, there has been a long-standing dispute between the James–Lange theory, which postulates that emotions follow initial peripheral physiological events as a central response, and other theories that postulate causality in the other direction[28,29]. In addition to other accumulating evidence, a landmark study in mice showed that a specific emotion (anxiety-like behaviour) developed through a reciprocal interaction between peripheral input — that is, an increase in heart rate induced by optogenetics — and brain activity in the posterior insular cortex[26]. Another animal study showed that the interrelationship between short-term neural and cardiac activities during emotions (so-called microstates) is shaped by longer-term conditions (so-called macrostates)[30]. The notion of macrostates has now been refined to refer to long-lasting, structurally entrenched neurobehavioural–cardiovascular (brain–body) states shaped and influenced by genetics, diet, environmental factors and exercise, and corresponding to determinants of cardiovascular and mental health[9]. Longer-term changes in these macrostates that couple cardiovascular and brain function are likely to have crucial pathophysiological roles in conditions such as depression, hypertension[31], post-traumatic stress disorder, TTS and SHS (see sections Takotsubo syndrome and Stroke–heart syndrome below).

Attempts are being made to identify clinically useful physiological measures of the integrity of heart–brain coupling[32]. Simultaneous electrocardiogram (ECG) and EEG recordings yield intriguing parameters of brain–heart interaction. Averaging the EEG to the R peak (or, alternatively, the T wave) of the ECG results in heartbeat-evoked potentials (HEPs)[33]. The dipole sources of the HEP have been localized in the anterior cingulate, right insular, prefrontal and left secondary somatosensory cortices, all of which are crucial for brain–heart communication[34]. HEPs have been shown to be altered in atrial fibrillation[35], hypertension[36] and frontotemporal dementia[37], among other conditions. Despite their promise as a biomarker, several important methodological and conceptual issues regarding the validity of HEPs must be considered before they can be applied in the clinic.

In recent years, several other highly promising methods for assessing brain–heart interactions have been developed and extensively studied[32]. These approaches can be categorized into: those that measure cortical responses to the heartbeat (beyond HEPs); those that identify neural correlates of cardiac rhythms by combining neuroimaging methods such as functional MRI (fMRI) and autonomic outflow measures such as heart rate variability (HRV); those that assess dynamic brain–heart oscillatory coupling; those that determine causal interactions by applying selective neuromodulation techniques such as optogenetics, TMS or deep brain stimulation to investigate how neural pathways influence heart function and vice versa; and those that apply methods to identify higher-order or multilayer interactions[32]. In parallel, marked advances have been made in decoding the brain–heart axis through the use of non-invasive brain, vasculature and heart imaging techniques, both individually and in combination (Box 1).

Emotional stress and cardiovascular disease

Mental stress-induced myocardial ischaemia

Emotional stress, both acute and chronic, increases the risk of cardiovascular disease. Acute stress can trigger mental stress-induced myocardial ischaemia (MSIMI) and TTS, and chronic stress, depression

Box 1 | Multimodal tools to study the brain–heart axis

Brain domain

Ultra-high-field structural MRI and diffusion tensor imaging (DTI) at submillimetre spatial resolution allow exploration of pathways that link the hypothalamus, insular cortex, amygdala and prefrontal cortex. Resting-state or task functional MRI (fMRI) maps their functional connectivity during autonomic or emotional challenges. Magnetic resonance spectroscopy (MRS) quantifies neurochemical excitatory and inhibitory metabolites such as glutamate and GABA that are expressed in the brain. Cerebral PET allows quantification of perfusion, metabolism (using the tracer ^{18}F-fluorodeoxyglucose), neuroinflammation and sex hormone receptors. Newly developed long-axial field-of-view (LAFOV) scanners enable dynamic protocols that capture simultaneous brain and peripheral organ crosstalk.

Heart domain

Multiparametric cardiovascular MRI provides information on ventricular volumes, deformation, quantitative perfusion and tissue characteristics — most notably through T1 and T2 mapping and late gadolinium enhancement — thereby linking contractile reserve, microvascular state and tissue composition to central autonomic drive. Cardiac PET quantifies absolute myocardial blood flow, flow reserve, and sympathetic and parasympathetic innervation.

Vessel and microcirculation domain

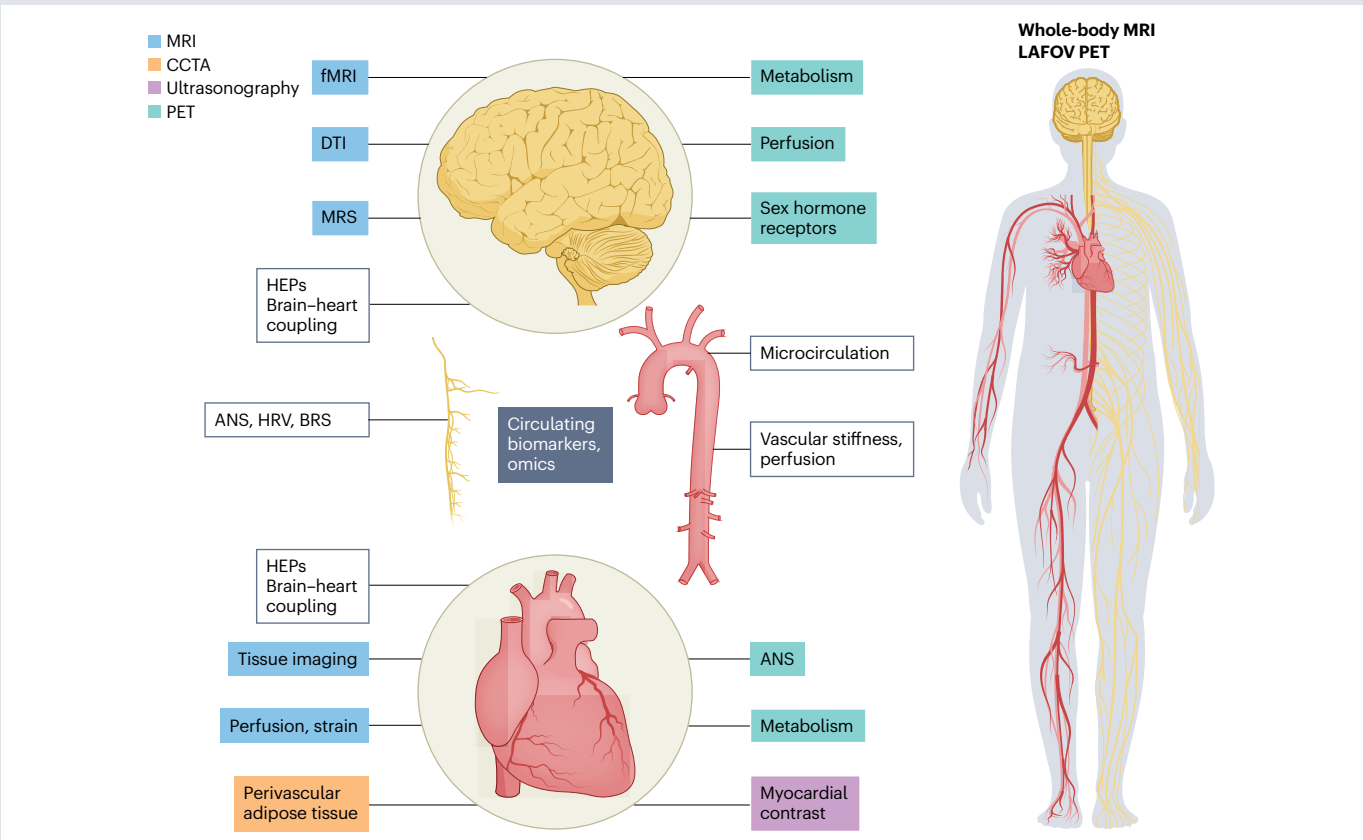
Cardiac PET and cardiovascular MRI measure cerebral and myocardial microvascular flow, and CT-derived pericoronary fat attenuation indexes perivascular inflammation. Together, these modalities bridge small vessel disease (SVD) in the brain and heart with systemic inflammatory tone.

Whole-body perspective

Using multiorgan MRI and LAFOV PET scanners (right-hand panel in the figure) interorgan time–activity curves can be acquired within a single examination, enabling integrated pharmacokinetic or stress–challenge paradigms. Whole-body imaging can be merged with circulating biomarkers, multiomics and wearable sensors to unravel biological and functional pathways involved in brain–heart crosstalk.

Study designs and translational potential

Time-locked mental stress, baroreflex or drug challenges can be synchronized with high temporal resolution PET and MRI to explore causal links between neural activation, autonomic outflow, myocardial perfusion and vascular tone. This framework supports discovery of sex-specific mechanisms, identification of early microvascular dysfunction that precedes neurodegeneration, and monitoring of



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interventions that target autonomic imbalance or inflammation. Ultimately, this multimodal strategy could shift heart–brain–vessel research from an organ-based to an integrated approach.

ANS, autonomic nervous system; BRS, baroreflex sensitivity; CCTA, coronary CT angiography; HEPs, heartbeat-evoked potentials; HRV, heart rate variability.

and anxiety are associated with an increased risk of major adverse cardiovascular events (MACE)[38–41]. Acute and chronic emotional stress can lead to behavioural changes, such as poor diet, smoking and lack of exercise, as well as physiological changes, including increases in heart rate, blood pressure and cortisol levels[42,43].

MSIMI is a phenomenon whereby psychological stress causes myocardial ischaemia in patients with coronary artery disease[44]. MSIMI occurs independently of heart disease severity and is associated with an increased risk of MACE[40]. Risk factors for MSIMI include female sex, younger age and chronic psychosocial stress[38,42]. The mechanisms underlying MSIMI include abnormal microvascular reactivity and inflammation[38,40,45]. In addition, brain fMRI studies have shown that in experimental stress paradigms, people with MSIMI and stress-induced peripheral vasoconstriction have increased brain activity, especially in regions that modulate the cardiac autonomic nervous system (ANS) such as the anterior cingulate and insular cortices[38,46]. More generally, the high co-dependence of mental and cardiovascular health has been conceptualized as reflecting tightly integrated neurocardiovascular brain–body states[9].

Takotsubo syndrome

Triggers and clinical features. TTS represents a textbook case of brain–heart interplay in which an acute cerebral disorder triggers cardiac dysfunction (Fig. 2). Also known as 'broken heart syndrome', TTS is typically triggered by intense emotional stress and shows hallmark features of transient left ventricular dysfunction and apical ballooning, leading to chest pain or dyspnoea[39]. Although TTS might be considered an extreme form of MSIMI, it is increasingly recognized to also occur secondarily to physical stress, especially acute neurological disorders such as seizures and stroke[47]. Contemporary studies even suggest a higher proportion of physical triggers than emotional triggers for TTS, possibly related to the increased awareness of this condition[48].

One of the most intriguing aspects of TTS is the striking influence of sex and gender on its epidemiology, clinical course and potential mechanisms[49,50]. TTS predominantly affects postmenopausal women, who account for approximately 85–90% of cases – a statistical finding that suggests a potential role for hormonal changes in its pathophysiology[48,49]. In addition, genetic, cultural and environmental factors might have roles in its development. In Japan, for instance, the proportion of people with TTS who are male is >10% higher than in European populations[51]. Compared with their female counterparts, men with TTS are younger, more often have physical rather than emotional triggers, and exhibit a more severe clinical course characterized by higher in-hospital mortality rates, necessitating close monitoring during hospitalization and long-term follow-up[52,53].

The outcomes of TTS seem to be related to the underlying trigger: in-hospital adverse cardiac events such as cardiac arrest or cardiogenic shock were found to occur in 4% of patients with TTS secondary to emotional stress and in up to 24% of patients with TTS secondary to neurological disorders[54]. In approximately 3% of patients with TTS, severe left ventricular wall motion abnormalities result in thrombus formation, which can lead to ischaemic stroke and systemic embolism[55]. In rare individuals with few cardiac symptoms who present with neurological deficits, it can be difficult to distinguish whether stroke was a cause or a consequence of TTS.

Although TTS has traditionally been considered benign, the long-term mortality risk attributable to cardiovascular and non-cardiovascular causes is higher than in the general population[56]. In this context, cardiovascular MRI (CMR) can yield prognostic information through characterization of myocardial tissue[57,58]. CMR can detect subtle long-term abnormalities such as impaired myocardial deformation and diffuse myocardial fibrosis at a median of 20 months after TTS, which could partly explain the persistence of clinical symptoms and long-term risk of MACE[58].

Pathophysiology and sex differences. Despite over three decades of research, the mechanisms underlying TTS remain poorly understood. The brain–heart axis is an important area of interest in TTS research, as it highlights the interplay between neuronal stress responses and cardiac function. Sex and gender differences in stress perception and autonomic regulation are believed to contribute to the higher prevalence of TTS in women than in men[50,53,59,60]. Neuroimaging studies suggest morphological and functional differences in brain regions of the CAN and limbic network (insular cortex, amygdala, cingulate cortex and hippocampus) between people with TTS and healthy controls[61–70] (Table 1). Of these regions, the amygdala, a key brain region involved in processing emotions and regulating sympathetic activity, exhibits sexual dimorphism. These sex differences include larger amygdala volumes in men, decreasing metabolic activity with age in men but not in women, and differences in hormone receptor expression[71–74]. Socioeconomic and lifestyle factors, which differ between women and men, also influence amygdala activity and might contribute to the observed sex differences in TTS[75,76]. The impact of the amygdala extends beyond TTS[63,65,68], influencing progression of atherosclerosis and adverse cardiovascular outcomes[77,78]. Studies suggest that increased amygdala activity is more strongly linked to inflammation and cardiovascular disease in women than in men[79–85].

Sex differences in both cardiac morphology and cardiac sympathetic activity that are observed with ageing probably have a role in TTS. Women have smaller ventricular dimensions and left ventricle mass but higher left ventricular systolic function than men. With increasing age, women have lower left ventricular end-diastolic volumes and a higher prevalence of heart failure with preserved ejection fraction compared with men[86–88]. Moreover, evidence suggests that ageing is related to sex-specific changes in regional cardiac sympathetic activity, as indicated by enhanced ^{18}F-DOPA uptake in the apical region of the left ventricle, which is particularly affected in TTS[59]. Hyperactivation of the SNS is considered to be a hallmark of TTS[89,90], as it leads to an increase in noradrenaline and adrenaline release, which triggers coronary vasoconstriction, microvascular dysfunction with subsequent acute myocardial

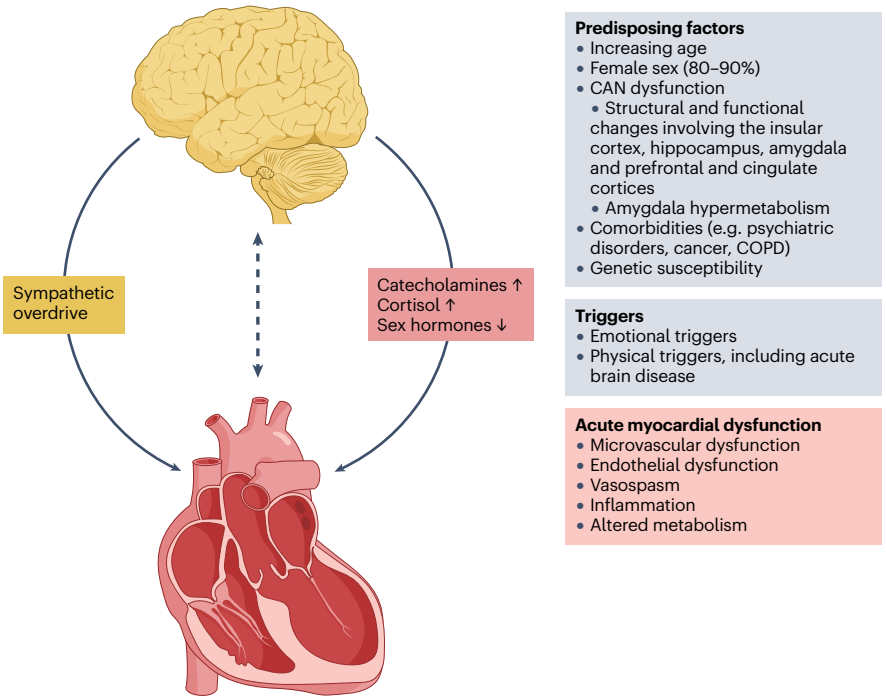

Fig. 2 | Brain–heart crosstalk in Takotsubo syndrome. In susceptible individuals with predisposing factors, emotional or physical stress triggers sympathetic overdrive and increased release of stress hormones, leading to myocardial dysfunction. CAN, central autonomic network; COPD, chronic obstructive pulmonary disease.

ischaemia and myocardial dysfunction[91,92]. At present, the effectiveness of beta blockers in mitigating these effects remains unclear. Observational studies have shown neutral effects of these drugs on short-term and long-term mortality, and no randomized controlled trials have been conducted to evaluate their efficacy[93,94]. A study published in 2025 also questions the efficacy of beta blocker treatment in female patients with acute coronary syndrome and preserved left ventricular function[95].

Sex hormones, particularly oestrogen, are believed to have a pivotal role in the development of TTS[96,97]. The condition is most prevalent in postmenopausal women, suggesting that a decline in oestrogen levels could contribute to increased susceptibility. Oestrogen has been shown to modulate β_1 and β_2 adrenergic receptor expression and to attenuate catecholamine-mediated vasoconstriction and reduce stress-induced cardiac dysfunction in rodent stress models, in human-induced pluripotent stem cell-derived cardiomyocytes and following mental stress paradigms in postmenopausal women[98–100]. Moreover, oestradiol – the main form of oestrogen produced by the ovaries before menopause – enhances parasympathetic tone, mitigates vascular inflammation and regulates blood pressure responses to mental stress while modulating the hypothalamic–pituitary–adrenal (HPA) axis[101,102]. Although observational studies have suggested that premenopausal reproductive factors can influence the later onset of TTS, no protective effect of hormone replacement therapy has been demonstrated, raising questions about the precise role of oestrogen in the syndrome[103,104]. The difficulty of assessing the effects of oestrogen might also stem from its naturally low levels in postmenopausal women, which complicates the interpretation of its influence on cardiovascular health[105].

Inflammation, both systemic and in the heart itself, is another recognized feature of TTS[106–111]. Sex differences in the immune system seem to be relevant, with women exhibiting higher baseline inflammatory markers and stronger pro-inflammatory responses to stress than men[78,112–114]. These factors contribute to a greater susceptibility to mental stress-induced endothelial dysfunction and myocardial ischaemia[115,116]. Acute stress results in prominent mobilization of pro-inflammatory leukocyte populations in animal models[117]. Further preclinical studies have revealed sex-specific differences in inflammatory pathways, with calcineurin signalling showing more pronounced upregulation in male than in female mice in an adrenaline-induced TTS model[118]. Immunosuppressive agents such as cyclosporine A have shown promise in alleviating symptoms in animal models, and case reports suggest that patients with TTS who have inflammatory comorbidities could benefit from corticosteroid therapy, but clinical data remain sparse[118–120].

In conclusion, TTS is a complex condition influenced by sex differences in brain–heart communication, which include neural regulation of the stress response, sympathetic outflow to the heart, sex hormones and the inflammatory response. The efficacy of therapeutic interventions such as beta blockers, anti-inflammatory treatments and hormone replacement therapy should be evaluated through well-designed clinical trials that overcome the current limitations of animal models and heterogeneity of patient populations (Box 2), and should consider sex-specific outcomes.

Brain–heart interactions in stroke

Cardiac complications after stroke

The bidirectional interaction between the heart and the brain in the setting of stroke manifests clinically in many ways. Cardiac sources of embolism constitute a major cause of stroke[121] and, conversely, acute ischaemic brain injury can have detrimental effects on the cardiovascular system[6]. Subtle cardiac changes that might be clinically asymptomatic are detectable in around half of patients following hospitalization for acute stroke[122]. These changes include ECG evidence of disturbed repolarization, left ventricular systolic and diastolic dysfunction, and elevation of cardiac biomarkers such as cardiac troponin

and natriuretic peptides in the blood, which indicate myocardial injury or stretch[6,47,122,123]. In 10–20% of patients, these post-stroke cardiac changes manifest clinically as heart failure, TTS, coronary syndromes, arrhythmia or even sudden cardiac death[122,124].

Cumulative adverse events data from the Virtual International Stroke Trials Archive confirmed that 12% of patients with ischaemic stroke experienced at least one severe cardiac adverse event within 30 days of stroke diagnosis[125]. In line with earlier studies[124], the median time between stroke and incident adverse cardiac events was 2–3 days and the majority of events occurred within 10 days of the index stroke[125]. This peak corresponds to the early stroke-induced stress response, which we discuss in more detail in section Pathophysiology below. Depending on the manifestation, the hazard ratio for 90-day mortality was 1.6-fold to 2.4-fold higher than in patients without post-stroke cardiac complications[125]. This finding is in line with several other independent cohort studies, which showed a strong association between early post-stroke cardiac changes and unfavourable short-term outcomes, most notably increased mortality[124,126–128]. Importantly, evidence is growing that adverse cardiovascular events are not limited to the immediate aftermath of stroke. Patients with stroke in whom cardiac injury is detected early after the event are at increased risk of MACE during long-term follow-up, and are probably also at increased risk of developing cognitive impairment and dementia[129–131].

A similar burden of early cardiovascular complications has been reported following haemorrhagic stroke. In a pooled analysis of acute stroke trials that enrolled patients with haemorrhagic stroke, 13% of patients experienced severe adverse cardiac events within the first month after stroke[132]. In a further extensive analysis of medical records of primarily US-based health-care beneficiaries, patients with haemorrhagic stroke and early cardiac complications were at 49% increased risk of death and 35% increased risk of MACE within 5 years of follow-up[133].

Stroke–heart syndrome

Risk factors and clinical features. Older age and a history of heart disease are independently associated with cardiac complications after stroke and, in addition, stroke-related factors increase the likelihood of cardiac complications[6]. Even after conventional cardiovascular risk factors are taken into account, higher initial stroke severity, larger lesion size, right hemispheric lesions and specific lesion locations show robust associations with post-stroke cardiac sequelae[6,134,135]. In particular, stroke affecting the insular cortex has repeatedly been linked to all cardiac manifestations but especially post-stroke myocardial injury, TTS secondary to stroke, arrhythmia and death[135–138]. The insular cortex belongs to the CAN and is linked to both afferent cardiac interoception and efferent modulation of the cardiovascular response to acute stress[19,139,140].

These findings, together with the clear temporal pattern of post-stroke cardiac changes[124,125,141] and a wealth of animal data demonstrating that heart injury can be induced by experimental stroke[122,142–145], led to the proposal of an integrated pathophysiological concept known as SHS[6]. The SHS mechanistic framework considers a stroke as a specific event that triggers neurogenic heart disease. SHS is operationally defined as either new evidence of cardiac alterations or documented worsening of premorbid cardiac function diagnosed within 30 days after stroke[122].

Pathophysiology. Within the past few years, several animal studies have shed light on the role of altered brain–heart signalling in SHS pathogenesis. This altered brain–heart communication involves both 'hardwired' connections through autonomic nerves and humoral factors in the bloodstream (Fig. 3).

One line of evidence highlights the influence of the stress axis, which includes the ANS and the HPA axis[6,7,146]. Following stroke,

Table 1 | Brain imaging studies in Takotsubo syndrome

Study	Population	Imaging method	Findings
Pereira et al.[61]	4 women with previous TTS and 8 healthy controls	fMRI	Following cold exposure or Valsalva manoeuvre, differential activation of the insular cortex, amygdala and right hippocampus between TTS and control groups
Klein et al.[62]	20 women with previous TTS and 19 healthy controls	Multimodal MRI	TTS group showed structural and functional alterations in brain regions constituting the emotional–autonomic control system
Hiestand et al.[63]	20 women with previous TTS and 39 healthy controls	fMRI	TTS group showed morphological differences in the limbic network (insular cortex, amygdala, cingulate cortex and hippocampus)
Silva et al.[64]	8 women with previous TTS and 8 healthy controls	fMRI	TTS group showed increased connectivity in regions involved in emotional and autonomic control
Templin et al.[65]	15 women with previous TTS and 39 healthy controls	fMRI	TTS group showed hypoconnectivity of brain regions associated with autonomic functions and limbic system regulation
Dichtl et al.[66]	13 women within 72 h of TTS and 13 healthy controls	Multimodal MRI	TTS group showed volumetric and functional changes in frontal regions and right insular cortex, anterior cingulate cortex and amygdala
Suzuki et al.[67]	4 patients (3 women) on days 2–4 and days 29–40 after TTS	^{18}F-FDG PET	Amygdala metabolic activity higher in the acute phase than in the recovery phase of TTS
Radfar et al.[68]	104 individuals (86% with malignancy), of whom 41 (29 women) subsequently developed TTS, and 63 controls	^{18}F-FDG PET	Higher amygdala metabolic activity associated with an increased risk of TTS and earlier TTS
Santoro et al.[69]	1 woman, a few hours before TTS	^{18}F-FDG PET	Relative hypermetabolism in the right amygdala during the hyperacute phase of TTS
Khan et al.[70]	25 patients (24 women) 5 days after TTS and 25 matched controls	fMRI	TTS group showed volumetric differences in the limbic system and increased connectivity between the left insular cortex and limbic structures

^{18}F-FDG, ^{18}F-fluorodeoxyglucose; fMRI, functional MRI; TTS, Takotsubo syndrome.

Box 2 | From research to real-world benefits

The bidirectional communication pathways between the heart and the brain interact in a complex way. Future models need to enable selective modulation of neuronal, mechanical and humoral communication circuits between the two organs to dissect causal contributions to clinical manifestations and outcomes.

Progress in artificial intelligence and handling of big data will allow complex measurements of brain–heart coupling and its dynamic fluctuations and their influence on neuropsychiatric phenotypes. These advances also hold promise for integrating multimodal data from multiorgan imaging, genomic and proteomic analyses, which will enable the prediction of injury phenotypes in the brain and heart, as well as clinical disease trajectories.

One of the most pressing issues in Takotsubo syndrome research is the lack of representative animal models that incorporate both sexes, a broad range of age groups and stress paradigms that mimic real-world triggers. Clinical studies should strive towards larger, more diverse patient populations, including men, who have been under-represented in this research.

At present, no tools are available to predict phenotypes and clinical outcomes of stroke–heart syndrome (SHS) at the individual patient level. It will be crucial to unravel the plasma proteomic landscape and changes in the immune compartment to characterize biologically diverse SHS phenotypes and identify novel circulating biomarkers.

An ongoing challenging is to disentangle the effects of disturbed brain–heart communication and shared cardiovascular and other risk factors on the occurrence of SHS and cognitive impairment in patients with heart disease. The use of mouse models that resemble the situation in humans (for example, older mice and models that take comorbidities into account) will be a prerequisite for meaningful clinical translation.

autonomic outflow and reflex arcs are altered, resulting in overactivation of the SNS or downregulation of the parasympathetic nervous system. These alterations are mirrored by changes in markers of sympathovagal balance such as HRV or heart rate turbulence, impaired baroreflex sensitivity (BRS) and nocturnal non-dipping of heart rate[128,147–149]. Acute stroke-induced cardiac injury in mice was shown to be accompanied by neuronal activation in the sympathetic superior cervical ganglion, disturbance of catecholamine homeostasis with increased catecholamine levels in the blood and heart, and increased blood cortisol levels[143,150,151]. These effects could be mitigated by beta blocker treatment or resection of the superior cervical ganglion[150,152]. Increased ANS activation after stroke with local and systemic release of catecholamines and stress hormones might lead to coronary spasm, endothelial and microvascular dysfunction, altered Ca^{2+} handling and Ca^{2+} overload, arrhythmia, and cardiomyocyte necrosis[6].

Activation of the ANS and the HPA axis also results in rapid adaptations of the immune system, with mobilization of neutrophils and monocytes into the blood circulation from peripheral immune cell reservoirs including the bone marrow, spleen and gut. Several animal studies have shown that neutrophils and monocyte-derived macrophages with a pro-inflammatory profile infiltrate the heart following experimental stroke[142,144,152,153]. Of note, the extent of macrophage invasion correlates with left ventricular cardiac dysfunction[152]. This dysfunction is inhibited by blocking C–C chemokine receptor type 2, which is responsible for monocyte recruitment.

Of note, the recruitment of pro-inflammatory monocytes into the heart is not a transient short-term phenomenon. In both mice and humans, stroke-induced pro-inflammatory transcriptomic phenotype changes in monocytes and macrophages have been shown to persist for several weeks and to result in chronic monocyte infiltration into the heart, which is accompanied by chronic cardiac dysfunction and induction of myocardial fibrosis[142]. This process is causally linked to an early post-stroke surge of IL-1β into the bloodstream. Importantly, breakdown of the blood–brain barrier shortly after brain ischaemia promotes the release of immunoactive molecules, known as damage-associated molecular patterns (DAMPs), from injured cells in the brain, activated cerebral immune cells and/or endothelial cells into the blood circulation, leading to an acute systemic inflammatory response[154]. These DAMPs induce systemic activation of immune cells with release of pro-inflammatory cytokines and chemokines.

Several studies in mice have demonstrated that experimental stroke not only results in inflammation in the brain but also induces an increase in systemic and local cardiac pro-inflammatory signals, including IL-1β, IL-6 and monocyte chemotactic protein 1 (refs. 144,152,153). Besides stroke-induced local cardiac inflammation and induction of fibrosis, evidence suggests that these processes also affect the vasculature and increase the vulnerability of atherosclerotic plaques through IL-1β-mediated inflammasome activation[155]. Supporting this line of evidence, NLRP3 inflammasome activation was observed in the heart and atherosclerotic plaques following experimental stroke. Stroke-induced cardiac dysfunction was rescued in NLRP3-deficient mice[152] or following NLRP3 inhibition[145], suggesting a causal role for inflammasome activation in SHS. Release of nuclear DNA into the extracellular space following ischaemic stroke results in the formation of neutrophil extracellular traps that contribute to thrombosis and atherosclerosis[154]. Moreover, activated circulating innate immune cells interact with platelets, promoting inflammation-dependent activation of the coagulation system, which results in a hypercoagulable state[156].

A study in mice published in 2024 used a temperature and ultrasound-responsive brain-targeted drug delivery system to reduce local cerebral inflammation following experimental stroke[157]. Targeting inflammation not only resulted in reduced stroke lesion volumes but also led to marked amelioration of stroke-induced cardiac dysfunction. This finding suggests that neuroprotection could prevent SHS manifestation. Interestingly, pro-inflammatory cytokine release in the brain further activates the HPA axis and SNS[158], which again highlights the interdependence of the stress axis and the immune system in SHS pathophysiology. Well-designed interventional studies are required to disentangle this complex interplay and translate these promising targets into clinical trials.

Stroke-related perturbations of interorgan communication further potentiate the pathophysiological changes outlined above (Fig. 3). As already mentioned, the immune organs are richly innervated by peripheral autonomic nerves that project directly to neurocircuits in the brainstem, amygdala, insular cortex and hypothalamus that are involved in stress regulation[159]. Increased sympathetic tone following

experimental stroke results in increased release of pro-inflammatory monocytes and neutrophils from the bone marrow[160]. The spleen is another relevant immune cell reservoir. In an animal model of SHS, splenectomy protected against post-stroke cardiac dysfunction and fibrosis by reducing expression of pro-inflammatory factors and macrophage invasion into the heart[144]. This finding underlines the relevance of the bone marrow–spleen axis in SHS pathogenesis. Evidence also suggests that stroke results in gut dysbiosis, with changes in the gut microbiome as well as gut barrier leakage[161]. As a consequence, intestinal bacteria and immunoactive, neuroactive and vasoactive metabolites, such as short-chain fatty acids or trimethyl amine *N*-oxide, translocate into the blood, which further aggravates the systemic pro-inflammatory state after stroke, promotes vascular dysfunction and impairs the blood–brain barrier. Another organ that influences brain–heart crosstalk is the liver[162]. Following stroke and subsequent activation of the SNS and the HPA axis, hepatic glucose, energy and lipid metabolism are disrupted, leading to stress hyperglycaemia. Glutathione production is also increased, which might counterbalance the deleterious effects of oxidative stress after stroke. In addition, stroke and other types of brain injury affect the lungs, leading to pulmonary oedema, lung injury and increased vulnerability to infections[163]. Beyond the subsequent augmentation of neural, inflammatory and neuroendocrine pathways, impaired oxygenation of the blood resulting from hypoxia can lead to coronary demand ischaemia with myocardial damage and dysfunction[164].

An additional important interorgan communication pathway that is relevant to SHS and other systemic consequences after stroke is the release of EVs by injured CNS cells[7,16,165]. EVs are loaded with signalling molecules such as cytokines and microRNAs, which are considered to mediate physiological communication between cells[16]. In the setting of brain injury, activated microglia and damaged microvascular endothelial cells show enhanced secretion of EVs, which show alterations in pro-inflammatory content[16]. Importantly, EVs can cross the blood–brain barrier and gain access to the systemic circulation. Therefore, secreted EVs and their bioactive cargo not only affect the local cerebral microenvironment following stroke but also reach and affect the immune system, coagulation system and peripheral organs, including the heart[16].

In summary, acute ischaemic brain injury profoundly alters brain–heart signalling and interorgan communication[7], which can manifest as cardiac injury and dysfunction and strongly influences clinical outcomes. Although robust evidence indicates that disturbed

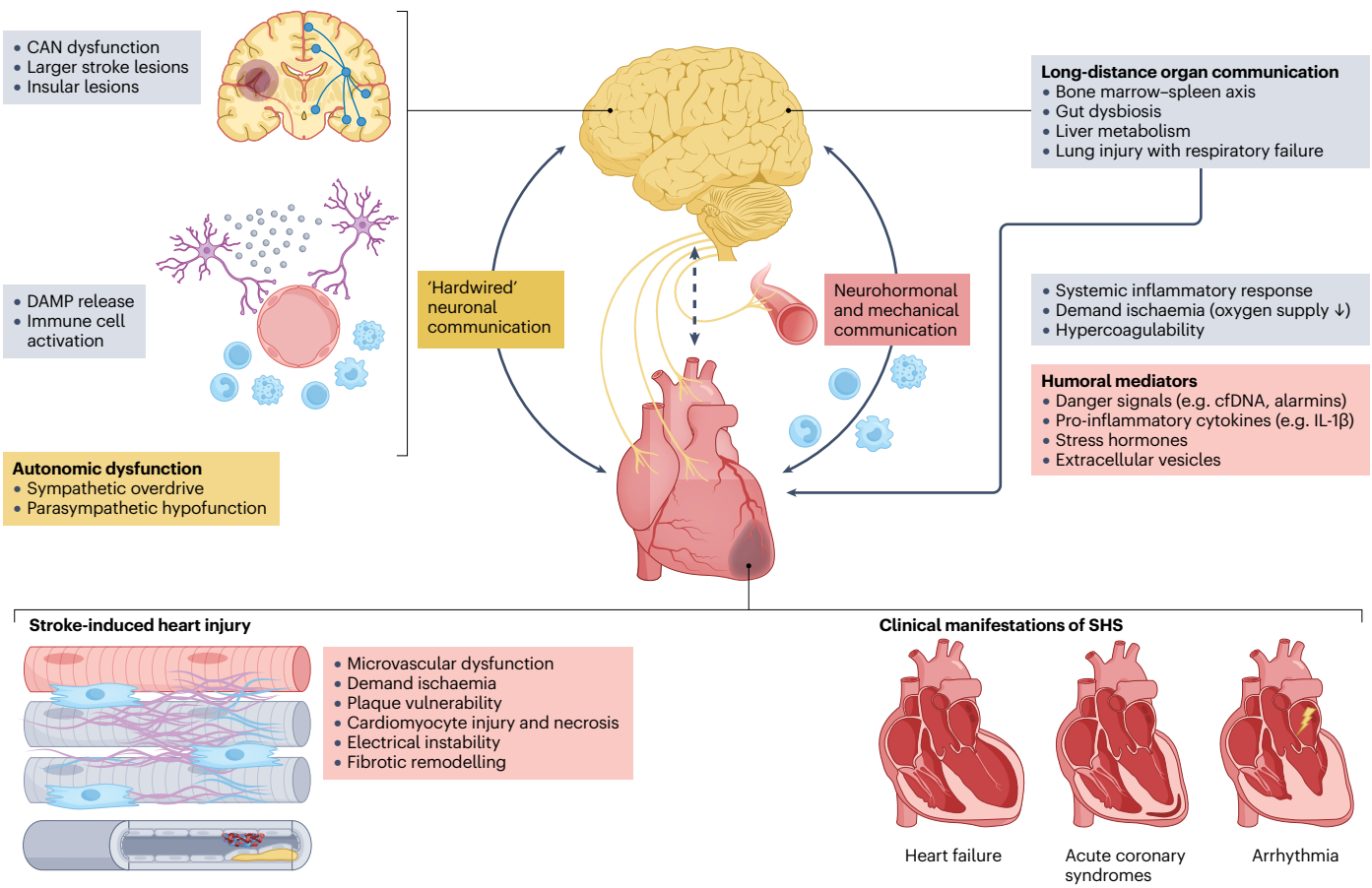

Fig. 3 | Brain–heart crosstalk in stroke–heart syndrome. Stroke, especially large lesions and lesions with insular cortex involvement, leads to changes in the central autonomic network (CAN) and breakdown of the blood–brain barrier with release of damage-associated molecular patterns (DAMP). As a consequence, autonomic dysfunction with sympathetic overdrive and an increased systemic inflammatory response lead to stroke-induced heart injury and, potentially, clinically relevant cardiac events. Other interorgan pathways involving the bone marrow, spleen, gut, liver and lungs amplify these processes. cfDNA, cell-free DNA; SHS, stroke–heart syndrome.

interactions between the ANS, the HPA axis and the immune system are paramount in SHS pathophysiology, we still lack evidence-based therapeutic interventions. Well-designed studies are needed to test potential approaches to prevent SHS and long-term cardiovascular events.

Sex and gender differences. Sex differences are evident in SHS but are much less pronounced than in TTS[135,166,167]. Overall, the global incidence of SHS manifestations is similar between men and women[168]; however, high troponin levels, acute myocardial injury and TTS following stroke are more common in women than in men[127]. Also, woman have a higher risk of death or poor outcome of acute myocardial injury after stroke[169]. Owing to differences in inflammation, sympathetic tone and stress responses, women might be at increased risk of stroke-related cardiac complications[170,171]. Although some studies have found no significant sex differences in MACE after stroke, a large UK study found a higher incidence of stroke and subsequent cardiovascular complications in women than in men[172]. The PRAISE (Prediction of Acute Coronary Syndrome in Acute Ischaemic Stroke) study analysed patients with stroke who had troponin elevation that fulfilled the rule-in criteria for acute coronary syndromes. Similar numbers of men and women were found to fulfil these criteria after ischaemic stroke, but the men were more likely to have atherothrombotic (type 1) myocardial infarction rather than stress-induced troponin elevation[173]. As such findings have implications for triaging the need and timing of invasive cardiac diagnostics after stroke, future studies on the mechanisms and outcomes of SHS should focus more on potential sex and gender differences.

Heart disease and cognitive impairment

Epidemiology

Growing evidence implicates a direct brain–heart link in the development of cognitive impairment and dementia. Although studies are often difficult to compare owing to methodological differences in neuropsychological testing, lack of data on the real-life impact of cognitive measures, and different clinical settings (for example, acute inpatient versus outpatient), solid evidence points towards a high burden of cognitive impairment in patients with manifest heart disease[8].

In patients with established heart disease, including myocardial infarction, atrial fibrillation and heart failure, cross-sectional studies have shown an increased prevalence of cognitive impairment, and longitudinal studies have shown an increased incidence of cognitive impairment and dementia[174–177]. Although most studies did not test different cognitive domains, myocardial infarction, atrial fibrillation and heart failure have been consistently linked to a decline in executive functions[174,178,179]. The strength of association between heart disease and incident dementia and cognitive decline might be age-dependent: in a UK Biobank study, the highest vulnerability was observed in individuals with coronary heart disease diagnosed before the age of 45 years[180]. In the population-based Whitehall II study, longer duration of atrial fibrillation was associated with faster cognitive decline and a higher risk of incident dementia, which was not entirely explained by incident stroke events[181]. In patients with newly diagnosed heart failure, however, cognitive decline was significantly steeper at older ages[175]. This association was independent of concomitant atrial fibrillation and heart failure phenotype (that is, preserved versus reduced ejection fraction).

Other studies have supported a dose–response correlation between severity of heart disease and severity of cognitive impairment. In patients with atherosclerotic coronary artery disease, increased severity of coronary microcirculatory dysfunction, as measured by coronary flow reserve, was associated with worse cognitive test scores[182]. Although subclinical, device-detected – that is, low-burden – atrial fibrillation was not associated with change in global cognitive function over a 3-year period[177], evidence suggests that patients with non-paroxysmal – that is, persistent or permanent high-burden – atrial fibrillation show lower global cognitive function than those with paroxysmal atrial fibrillation[183]. In addition, a dose-dependent correlation has been observed between prevalence of cognitive impairment and severity of heart failure, as measured by the New York Heart Association classification[184].

Even 'silent' heart disease, such as atrial cardiopathy without established atrial fibrillation, or echocardiography-detected impairment of myocardial deformation without clinically overt heart failure, has been consistently linked to cognitive impairment in both cross-sectional and longitudinal studies[185,186]. Moreover, in otherwise healthy community-dwelling adults and patients with recent stroke, levels of biomarkers of myocardial damage, such as high-sensitivity cardiac troponin or amino-terminal pro-B-type natriuretic peptide, have been shown to correlate with reduced attention and executive function, structural brain changes including brain network segregation, and incident – especially vascular – dementia[131,187–189].

Potential mechanisms

A pathophysiological link between dementia and heart disease was first proposed many decades ago. In an editorial in *The Lancet* in 1977, the term 'cardiogenic dementia' was coined following observations of "a mild dementing state [that] has become much more severe in the presence of physical illness such as … heart-failure"[190]. Several interacting mechanisms have been suggested to underlie cognitive impairment in patients with heart disease (Fig. 4). Although shared risk factors might damage both organs, as well as the vascular system, the relevance of brain–heart crosstalk is increasingly recognized.

One final common pathway that mediates cognitive dysfunction in patients with heart disease is brain damage caused by cerebral embolism and progression of cerebral SVD. Silent brain infarcts and severe SVD are strongly associated with an increased risk of cognitive impairment (in particular, vascular cognitive impairment) and future dementia[191–193]. Neuroimaging studies reveal a high burden of cerebral pathologies in patients with heart disease[193–196], including cerebral infarcts, which are often silent and are especially common in people with atrial fibrillation, SVD or cortical atrophy in distinct brain regions (especially in people with heart failure). In a meta-analysis of hospital-based and community-based studies, more than one-third of individuals with heart disease (including myocardial infarction, atrial fibrillation and heart failure) had silent brain infarcts, nearly two-thirds had white matter hyperintensities and one-third had other markers indicative of SVD-like lacunes, cerebral microbleeds and enlarged perivascular spaces[194]. The findings varied by age and history of hypertension but not by history of stroke and not substantially by type of heart disease.

A subsequent study, however, indicated substantial differences in the prevalence of various SVD markers between patients with atrial fibrillation and atherosclerotic coronary artery disease[197]. The patients with atrial fibrillation had more embolic infarcts and periventricular white matter hyperintensities, whereas those with systemic atherosclerosis had more lacunar infarcts[197]. Given that periventricular white matter hyperintensities and other SVD changes differ in terms of histopathology, risk factors and secondary neurodegeneration[198–201], this finding supports the idea that distinct brain injury processes might be at play in patients with atrial fibrillation. Importantly, the association of cardiac dysfunction and arrhythmia with markers of cerebral SVD

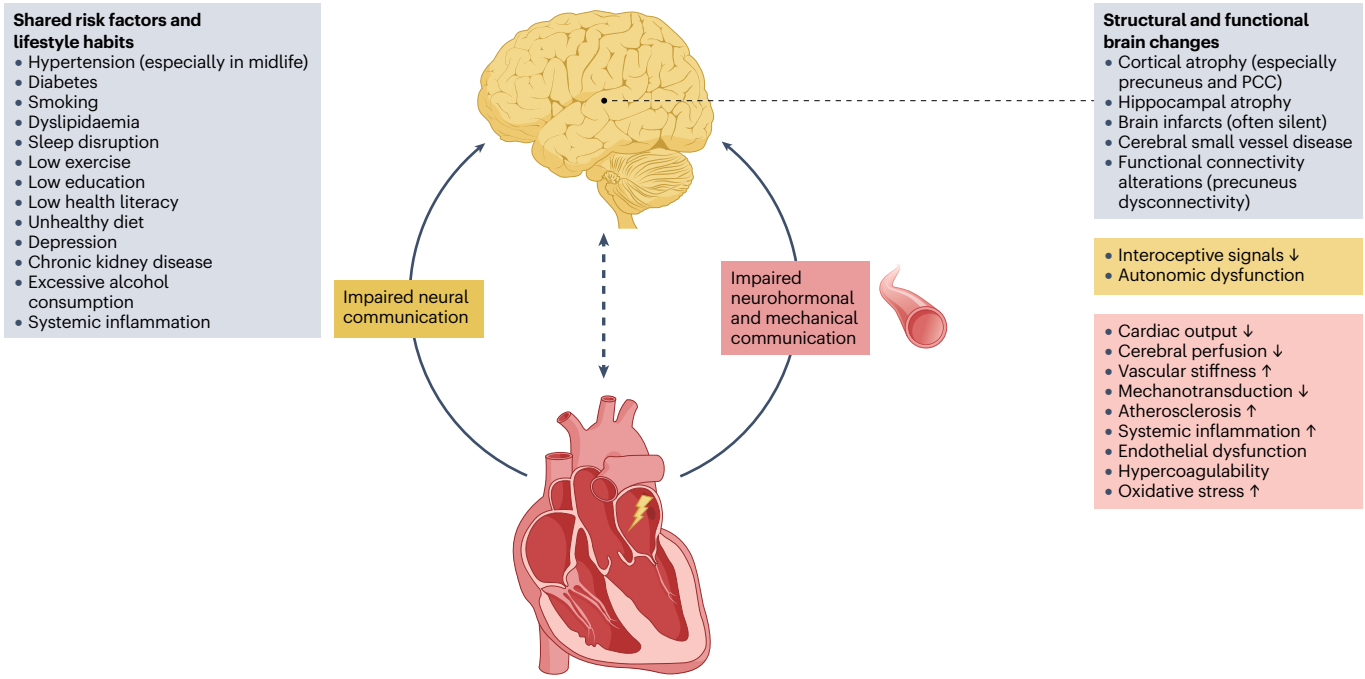

Fig. 4 | Cognitive impairment in patients with heart disease. Impairments in neural, neurohormonal and mechanical brain–heart signalling result in structural and functional brain changes with subsequent cognitive impairment (key drivers shown in red and yellow boxes). Risk factors (grey box) acting on the brain and cardiovascular system aggravate these processes. PCC, posterior cingulate cortex.

and incident dementia is independent of cardiovascular risk factors and history of stroke[178,202,203], suggesting that it is not mediated solely by shared cardiovascular risk factors.

Beyond cerebral embolism and SVD, evidence exists of distinct patterns of brain atrophy in patients with heart disease, especially those with heart failure. In one study, compared with healthy control individuals, patients with mild and stable heart failure exhibited atrophy of the medial temporal lobe encompassing the hippocampus and the amygdala[204]. This difference correlated with cognitive impairment and persisted during a 3-year follow-up period but did not progress[205]. In another study, patients with coronary heart disease and markers of heart failure showed structural changes in the medial and posterior cingulate cortex, precuneus and hippocampus, which correlated with heart failure severity[195]. In addition to these structural changes, evidence exists that patients with heart failure have reduced connectivity between the precuneus and other widespread brain regions. Brain fMRI revealed that weaker precuneus connectivity correlates with markers of heart failure severity and with impaired cognitive performance[206]. The role of the hippocampus in memory functions is well established[207], and the precuneus and posterior cingulate cortex belong to the default mode network, which has been shown to be disrupted in patients with Alzheimer disease (AD)[208]. Moreover, the precuneus and cingulate cortex share strong connections to CAN structures such as the insular cortex and are considered to be an integrating area for interoceptive information and contextual information of emotional states[209].

There is increasing interest in cardiac interoceptive signals and the associated regulation of the body's response (allostasis) for their possible pathophysiological roles in the development of various neurodegenerative disorders, including AD[210,211]. Interoceptive signals from mechanoreceptors, thermoreceptors and chemoreceptors in the heart and the vascular system reach the brain through the ANS and mechanoreceptive neurons that transduce arterial shear stress and blood pressure pulsations into neuronal signals[17,18,212]. These cardiac interoceptive signals are integrated within an allostatic interoceptive brain network, which overlaps with the CAN and includes the anterior cingulate cortex, insular cortex and dorsal amygdala[211]. Patients with heart disease exhibit impairments in ANS signalling, which are mirrored by impaired BRS and reduced HRV[213,214]. Myocardial infarction can lead to direct ischaemic damage to cardiac autonomic nerves, and haemodynamic changes in people with myocardial infarction, atrial fibrillation or heart failure can alter neurohumoral activity, with increased sympathetic activation and persistent maladaptations within the ANS[215]. In the community-based Rotterdam Study, reduced BRS was associated with an increased risk of incident dementia[216], and in the Multi-Ethnic Study of Atherosclerosis, reduced HRV correlated with impaired cognitive performance[217].

HEPs, which represent EEG-derived measures of the cortical representation of heartbeats, have emerged as a marker of neural brain–heart communication that modulates several cognitive processes. Interestingly, HEP amplitudes, especially in the insular cortex, were shown to be attenuated in patients with atrial fibrillation[35]. This finding might mirror attenuated cardiac interoception in atrial fibrillation, in which the heart beats irregularly and randomly. Of note, neuronal discharge in the hippocampus is modulated by the cardiac cycle, and the discharge rate correlates with changes in heart rate[218]. Altered brain–heart connections in people with cardiac arrhythmia might contribute to cognitive dysfunction and behavioural symptoms of dementia, such as impaired decision-making. This idea is supported

by the observation that the risk of incident dementia decreases in patients with atrial fibrillation who have undergone catheter ablation with restoration of sinus rhythm[219]. Collectively, altered interoceptive signals related to heart disease (for example, due to arrhythmia or reduced cardiac output) and aberrant patterns of ANS responses could result in allostatic interoceptive overload, leading to cognitive and behavioural problems[211,220].

Arterial stiffness and distensibility can affect physiological neuromechanical brain–heart communication. In a cross-sectional analysis of cerebral and CMR data from over 30,000 middle-aged UK Biobank participants, most of whom were healthy, a robust association was found between adverse brain MRI phenotypes, such as reduced grey matter volume, and cardiac imaging-derived phenotypes, including impairments in left ventricular function and aortic distensibility[221]. The strongest association was with reduced aortic distensibility, which correlates with aortic stiffness. These associations remained statistically significant after adjustment for vascular risk factors and lifestyle factors. Better cognitive performance measured in fluid intelligence and reaction time tests was associated with favourable CMR traits, including larger left and right ventricular volumes and greater aortic distensibility[222].

Another proposed mechanism underlying the link between heart disease and cognitive impairment is cerebral hypoperfusion. Hypoperfusion damages the neurovascular unit through energy imbalance, oxidative stress and inflammation[192,223]. Several brain regions with crucial roles in memory processes, such as the hippocampus, are particularly vulnerable to hypoperfusion[224]. Perfusion MRI has shown that atrial fibrillation is associated with reduced cerebral blood flow[225]. Compared with healthy controls, patients with severe heart failure had a 31% reduction in PET-derived measures of cerebral blood flow, which improved after heart transplantation[226]. This improvement was attributed to augmentation of cardiac output. Consistent with a pathophysiological role for hypoperfusion in cognitive impairment, global measures of cognition improved in around one-third of patients with heart failure within 8 months of implantation of a left ventricular assist device[227].

The contribution of systemic inflammation to the pathogenesis of heart failure, atrial fibrillation and atherosclerotic coronary artery disease is well established[228,229]. As described in the section Stroke–heart syndrome above, organ damage, including various types of cardiac injury, results in release of DAMPs with subsequent activation of the innate immune system[154,228]. Acute systemic inflammation can cause immediate brain injury and cognitive dysfunction, and chronic systemic inflammation with increased circulating pro-inflammatory cytokines can damage the neurovascular unit and increase the risk of cognitive decline[230,231]. Systemic inflammation might be 'sensed' by brain regions of the CAN, including the insular cortex, which affect neural outputs to the heart and potentially aggravate ANS dysfunction[232]. Of note, neuroinflammation with pro-inflammatory cytokine expression by microglial cells was detected in a rodent model of myocardial infarction with subsequent heart failure and reduced ejection fraction. This response was mediated by cardiac EVs abundant in miRNA-21-5p — a microRNA that promotes cardiac fibrosis and maladaptive cardiac remodelling[233]. Similarly, EVs released from the epicardial adipose tissue in a canine model of atrial fibrillation were shown to cross the blood–brain barrier and activate microglia in the hippocampus[234]. However, whether this mechanism of brain–heart crosstalk is relevant for cognitive decline in humans remains to be determined.

Overall, evidence suggests that altered bidirectional brain–heart feedback loops, impaired cerebral perfusion and inflammation can lead to vascular and microvascular brain damage and contribute to the high burden of cognitive dysfunction in patients with heart disease. Of note, similar mechanisms, including CAN dysfunction with altered sympathovagal dynamics, have been suggested to contribute to other sequelae of heart disease such as depression and anxiety[235,236]. Targeting these mechanisms and preventing heart disease and heart disease progression should contribute to preserving brain health in our ageing population.

Sex and gender differences

Evidence suggests that sex disparities in the brain–heart interaction are relevant to cognitive dysfunction in people with heart disease[237]. A prospective analysis of 464,616 UK Biobank participants found that among those who experienced a cardiovascular disease event, women were 1.5 times more likely than men to develop AD but had a 15% lower risk of developing vascular dementia[238]. Women with heart failure, especially those with preserved ejection fraction, exhibit greater cognitive decline than their male counterparts, possibly due to cerebral perfusion impairments, microvascular dysfunction and systemic inflammation[182,239]. These effects are exacerbated by hormonal changes during menopause, which accelerate vascular ageing[240]. Heart failure with preserved ejection fraction, which predominantly affects women, is often associated with hypertension and diabetes, further contributing to endothelial dysfunction and cognitive impairment[202,239–241]. Of note, despite growing research into brain–heart interactions[202,242,243], sex and gender differences remain largely overlooked in neuroscience studies[244], as highlighted by criticisms of AD clinical trials[245].

Conclusions and implications for patient care

The dynamic interplay between the heart and the brain is crucial for physiological regulation in healthy individuals, and dysfunction in one of these organs can adversely affect the other. In recent years, our understanding of both the physiological and pathophysiological mechanisms underlying this relationship has improved. Overcoming the translational challenges of existing disease models (Box 2) will enable more effective treatment and, potentially, prevention of the clinical burden associated with brain–heart crosstalk in the future. A deeper understanding of brain–heart communication will have implications for patient care across various clinical scenarios and potentially guide the development of diagnostic and prognostic tools. For instance, integrating measures of brain–heart coupling derived from multimodal monitoring in intensive care units could guide prognostic decision-making in people with stroke or other types of brain injury[128,246]. Improving brain–heart communication through neurofeedback and implementing strategies to reduce the allostatic load, in particular, the interoceptive allostatic load, might alleviate emotional stress in people with neuropsychiatric disorders, such as anxiety, and could also enhance cognitive function and psychological well-being[211,247]. The concepts derived from research into TTS and stroke–heart interactions that we describe in this Review will advance our overall understanding of brain–heart interactions. A broader view of brain–heart interactions and increasing clinical awareness of neurogenic heart disease has implications for the care of patients with other acute brain disorders, including seizures, brain haemorrhage and traumatic brain injury.

Collectively, the evidence outlined in this Review underscores the urgent need to foster neurocardiology collaboration beyond traditional boundaries. In this regard, a crucial challenge is the fragmented care of patients that promotes thinking in silos. Patients are often assessed and treated separately by distinct specialties, leading to missed opportunities

for comprehensive management. The establishment of joint clinical neurocardiology services and multidisciplinary research institutions and infrastructures with links to systems physiology, genetics and epigenetics, immunology and psychiatry should become a priority. Offering integrated patient-centred care pathways with multidisciplinary teams has been shown to improve long-term outcomes in patients with stroke or atrial fibrillation and is advocated by the latest cardiology guidelines[248,249]. Currently, however, these concepts are not widely applied.

Collaborative efforts are also relevant from a research perspective. Clinical trials in stroke and other areas of neurology should gather more information about cardiac complications and cardiac symptoms and, conversely, cognitive end points and comprehensive measures to determine overall brain health should be integrated into cardiovascular trials. Given the sex-specific mechanisms of brain–heart interactions, sex and gender differences should also be considered. Recognizing the complex interplay between central and peripheral neuronal function, systemic inflammation, microvascular regulation, hormonal changes and cardiovascular risk factors will be crucial for developing tailored interventions for prevention of cardiovascular and cerebrovascular events. Advances in our understanding of neurovascular and neurocardiac interfaces should aid the development of novel treatments to prevent cardiac dysfunction and progression of atherosclerotic cardiovascular disease[155,250]. Targeting of sympathetic overactivity and inflammation is a promising approach to reduce the burden of TTS and SHS, but we still lack specific therapeutic interventions and the therapeutic time windows are unknown.

At present, physicians should focus on controlling common risk factors that affect brain–heart communication (Fig. 1) and supporting adherence to established preventive approaches, including those that target the prevention of cognitive dysfunction in patients with heart disease (Fig. 4). Of note, preservation of cardiac function from midlife onward is associated with reductions in brain atrophy and neurodegeneration over time[251,252]. It is important to emphasize prevention by starting to screen for risk factors long before the clinical manifestation of brain or heart disease. In summary, the clinical relevance of bidirectional brain–heart interactions and their sex-specific differences, as we describe in this Review, strongly indicates that efforts to preserve brain health will prevent cardiac disease and vice versa.

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References

1. GBD 2021 Nervous System Disorders Collaborators Global, regional, and national burden of disorders affecting the nervous system, 1990-2021: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet Neurol.* **23**, 344–381 (2024).
2. Martin, S. S. et al. 2024 heart disease and stroke statistics: a report of US and global data from the American Heart Association. *Circulation* **149**, e347–e913 (2024).
3. GBD 2021 Forecasting Collaborators Burden of disease scenarios for 204 countries and territories, 2022-2050: a forecasting analysis for the Global Burden of Disease Study 2021. *Lancet* **403**, 2204–2256 (2024).
4. Quadt, L., Critchley, H. D. & Garfinkel, S. N. The neurobiology of interoception in health and disease. *Ann. N. Y. Acad. Sci.* **1428**, 112–128 (2018).
5. Al, E. et al. Heart-brain interactions shape somatosensory perception and evoked potentials. *Proc. Natl Acad. Sci. USA* **117**, 10575–10584 (2020).
6. Scheitz, J. F., Nolte, C. H., Doehner, W., Hachinski, V. & Endres, M. Stroke–heart syndrome: clinical presentation and underlying mechanisms. *Lancet Neurol.* **17**, 1109–1120 (2018).
7. Chen, Z. et al. Brain-heart interaction: cardiac complications after stroke. *Circ. Res.* **121**, 451–468 (2017).
8. van Nieuwkerk, A. C. et al. Cognitive impairment in patients with cardiac disease: implications for clinical practice. *Stroke* **54**, 2181–2191 (2023).
9. Villringer, A., Nikulin, V. V. & Gaebler, M. Brain-body states as a link between cardiovascular and mental health. *Trends Neurosci.* **48**, 766–779 (2025).
10. Chapleau, M. W. in *Primer on the Autonomic Nervous System* 4th edn (eds Biaggioni, I. et al.) 171–177 (Academic, 2023).
11. Lucini, D., Norbiato, G., Clerici, M. & Pagani, M. Hemodynamic and autonomic adjustments to real life stress conditions in humans. *Hypertension* **39**, 184–188 (2002).
12. Mussalo, H. et al. Baroreflex sensitivity in essential and secondary hypertension. *Clin. Auton. Res.* **12**, 465–471 (2002).
13. Mahinrad, S., de Craen, A. J. M., Yasar, S., van Heemst, D. & Sabayan, B. Natriuretic peptides in the central nervous system: novel targets for cognitive impairment. *Neurosci. Biobehav. Rev.* **68**, 148–156 (2016).
14. Volpe, M., Gallo, G. & Rubattu, S. Endocrine functions of the heart: from bench to bedside. *Eur. Heart J.* **44**, 643–655 (2023).
15. Reichmann, F. & Holzer, P. Neuropeptide Y: a stressful review. *Neuropeptides* **55**, 99–109 (2016).
16. Hirsch, Y. et al. Unpacking the role of extracellular vesicles in ischemic and hemorrhagic stroke: pathophysiology and therapeutic implications. *Transl. Stroke Res.* **14**, 146–159 (2023).
17. Jammal Salameh, L., Bitzenhofer, S. H., Hanganu-Opatz, I. L., Dutschmann, M. & Egger, V. Blood pressure pulsations modulate central neuronal activity via mechanosensitive ion channels. *Science* **383**, eadk8511 (2024).
18. Valenza, G., Matic, Z. & Catrambone, V. The brain-heart axis: integrative cooperation of neural, mechanical and biochemical pathways. *Nat. Rev. Cardiol.* **22**, 537–550 (2025).
19. Benarroch, E. E. The central autonomic network: functional organization, dysfunction, and perspective. *Mayo Clin. Proc.* **68**, 988–1001 (1993).
20. Quadt, L., Critchley, H. & Nagai, Y. Cognition, emotion, and the central autonomic network. *Auton. Neurosci.* **238**, 102948 (2022).
21. Makovac, E., Thayer, J. F. & Ottaviani, C. A meta-analysis of non-invasive brain stimulation and autonomic functioning: implications for brain–heart pathways to cardiovascular disease. *Neurosci. Biobehav. Rev.* **74**, 330–341 (2017).
22. Graham, F. K. & Clifton, R. K. Heart-rate change as a component of the orienting response. *Psychol. Bull.* **65**, 305–320 (1966).
23. Wilkinson, M., McIntyre, D. & Edwards, L. Electrocutaneous pain thresholds are higher during systole than diastole. *Biol. Psychol.* **94**, 71–73 (2013).
24. Al, E. et al. Cardiac activity impacts cortical motor excitability. *PLoS Biol.* **21**, e3002393 (2023).
25. Garfinkel, S. N. et al. Fear from the heart: sensitivity to fear stimuli depends on individual heartbeats. *J. Neurosci.* **34**, 6573–6582 (2014).
26. Hsueh, B. et al. Cardiogenic control of affective behavioural state. *Nature* **615**, 292–299 (2023).
27. Dworkin, B. R. et al. Central effects of baroreceptor activation in humans: attenuation of skeletal reflexes and pain perception. *Proc. Natl Acad. Sci. USA* **91**, 6329–6333 (1994).
28. Pace-Schott, E. F. et al. Physiological feelings. *Neurosci. Biobehav. Rev.* **103**, 267–304 (2019).
29. Lange, C. G. & James, W. (eds). *The Emotions* Vol. 1 (Williams & Wilkins, 1922).
30. Signoret-Genest, J. et al. Integrated cardio-behavioral responses to threat define defensive states. *Nat. Neurosci.* **26**, 447–457 (2023).
31. Schaare, H. L. et al. Associations between mental health, blood pressure and the development of hypertension. *Nat. Commun.* **14**, 1953 (2023).
32. Candia-Rivera, D., Faes, L., De Vico Fallani, F. & Chavez, M. Measures and models of brain–heart interactions. *IEEE Rev. Biomed. Eng.* https://doi.org/10.1109/RBME.2025.3529363 (2025).
33. Schandry, R. & Montoya, P. Event-related brain potentials and the processing of cardiac activity. *Biol. Psychol.* **42**, 75–85 (1996).
34. Pollatos, O., Kirsch, W. & Schandry, R. Brain structures involved in interoceptive awareness and cardioafferent signal processing: a dipole source localization study. *Hum. Brain Mapp.* **26**, 54–64 (2005).
35. Kumral, D. et al. Attenuation of the heartbeat-evoked potential in patients with atrial fibrillation. *JACC Clin. Electrophysiol.* **8**, 1219–1230 (2022).
36. Yoris, A. et al. Multilevel convergence of interoceptive impairments in hypertension: new evidence of disrupted body–brain interactions. *Hum. Brain Mapp.* **39**, 1563–1581 (2018).
37. Migeot, J. A., Duran-Aniotz, C. A., Signorelli, C. M., Piguet, O. & Ibanez, A. A predictive coding framework of allostatic-interoceptive overload in frontotemporal dementia. *Trends Neurosci.* **45**, 838–853 (2022).
38. Bremner, J. D. et al. Brain correlates of mental stress-induced myocardial ischemia. *Psychosom. Med.* **80**, 515–525 (2018).
39. Ghadri, J. R. et al. International expert consensus document on Takotsubo syndrome (part I): clinical characteristics, diagnostic criteria, and pathophysiology. *Eur. Heart J.* **39**, 2032–2046 (2018).
40. Mehta, P. K., Sharma, A., Bremner, J. D. & Vaccarino, V. Mental stress-induced myocardial ischemia. *Curr. Cardiol. Rep.* **24**, 2109–2120 (2022).
41. Park, H. D. & Blanke, O. Heartbeat-evoked cortical responses: underlying mechanisms, functional roles, and methodological considerations. *Neuroimage* **197**, 502–511 (2019).
42. Ebong, I. A. et al. The role of psychosocial stress on cardiovascular disease in women: JACC state-of-the-art review. *J. Am. Coll. Cardiol.* **84**, 298–314 (2024).
43. Moazzami, K. et al. Cardiovascular reactivity to mental stress and adverse cardiovascular outcomes in patients with coronary artery disease. *J. Am. Heart Assoc.* **14**, e034683 (2025).
44. Vaccarino, V. et al. Mental stress-induced-myocardial ischemia in young patients with recent myocardial infarction: sex differences and mechanisms. *Circulation* **137**, 794–805 (2018).

45. Shah, A. S. et al. Autonomic reactivity to mental stress is associated with cardiovascular mortality. *Eur. Heart J. Open*. **4**, oeae086 (2024).
46. Shah, A. et al. Brain correlates of stress-induced peripheral vasoconstriction in patients with cardiovascular disease. *Psychophysiology* **56**, e13291 (2019).
47. Cammann, V. L. et al. Clinical correlates and prognostic impact of neurologic disorders in Takotsubo syndrome. *Sci. Rep.* **11**, 23555 (2021).
48. Schweiger, V. et al. Temporal trends in Takotsubo syndrome: results from the international Takotsubo Registry. *J. Am. Coll. Cardiol.* **84**, 1178–1189 (2024).
49. Templin, C. et al. Clinical features and outcomes of Takotsubo (stress) cardiomyopathy. *N. Engl. J. Med.* **373**, 929–938 (2015).
50. Wittstein, I. S. Why sex matters in Takotsubo syndrome. *J. Am. Coll. Cardiol.* **79**, 2094–2096 (2022).
51. Imori, Y. et al. Ethnic comparison in Takotsubo syndrome: novel insights from the international Takotsubo Registry. *Clin. Res. Cardiol.* **111**, 186–196 (2022).
52. Arcari, L. et al. Gender differences in Takotsubo syndrome. *J. Am. Coll. Cardiol.* **79**, 2085–2093 (2022).
53. Cammann, V. L. et al. Age-related variations in Takotsubo syndrome. *J. Am. Coll. Cardiol.* **75**, 1869–1877 (2020).
54. Ghadri, J. R. et al. Long-term prognosis of patients with Takotsubo syndrome. *J. Am. Coll. Cardiol.* **72**, 874–882 (2018).
55. Salamanca, J. et al. Intraventricular thrombus in Takotsubo syndrome: incidence, predictors, management, and prognosis. Insights from the RETAKO registry. *Int. J. Cardiol.* **423**, 132985 (2025).
56. Rudd, A. E. et al. Cardiovascular and noncardiovascular prescribing and mortality after 778 Takotsubo comparison with myocardial infarction and general population. *JACC Adv.* **3**, 100797 (2024).
57. Arcari, L. et al. Cardiac magnetic resonance in patients with Takotsubo syndrome: clinical correlates of T2 mapping. *Int. J. Cardiol.* **419**, 132716 (2025).
58. Scally, C. et al. Persistent long-term structural, functional, and metabolic changes after stress-induced (Takotsubo) cardiomyopathy. *Circulation* **137**, 1039–1048 (2018).
59. Burger, I. A. et al. Age- and sex-dependent changes in sympathetic activity of the left ventricular apex assessed by ^{18}F-DOPA PET imaging. *PLoS ONE* **13**, e0202302 (2018).
60. Hogarth, A. J., Graham, L. N., Mary, D. A. & Greenwood, J. P. Gender differences in sympathetic neural activation following uncomplicated acute myocardial infarction. *Eur. Heart J.* **30**, 1764–1770 (2009).
61. Pereira, V. H. et al. Central autonomic nervous system response to autonomic challenges is altered in patients with a previous episode of Takotsubo cardiomyopathy. *Eur. Heart J. Acute Cardiovasc. Care* **5**, 152–163 (2016).
62. Klein, C. et al. Takotsubo syndrome — predictable from brain imaging data. *Sci. Rep.* **7**, 5434 (2017).
63. Hiestand, T. et al. Takotsubo syndrome associated with structural brain alterations of the limbic system. *J. Am. Coll. Cardiol.* **71**, 809–811 (2018).
64. Silva, A. R. et al. Brain functional connectivity is altered in patients with Takotsubo syndrome. *Sci. Rep.* **9**, 4187 (2019).
65. Templin, C. et al. Altered limbic and autonomic processing supports brain–heart axis in Takotsubo syndrome. *Eur. Heart J.* **40**, 1183–1187 (2019).
66. Dichtl, W. et al. Functional neuroimaging in the acute phase of Takotsubo syndrome: volumetric and functional changes of the right insular cortex. *Clin. Res. Cardiol.* **109**, 1107–1113 (2020).
67. Suzuki, H., Takanami, K., Takase, K., Shimokawa, H. & Yasuda, S. Reversible increase in stress-associated neurobiological activity in the acute phase of Takotsubo syndrome; a brain ^{18}F-FDG-PET study. *Int. J. Cardiol.* **344**, 31–33 (2021).
68. Radfar, A. et al. Stress-associated neurobiological activity associates with the risk for and timing of subsequent Takotsubo syndrome. *Eur. Heart J.* **42**, 1898–1908 (2021).
69. Santoro, F., D'Apollo, R. & Brunetti, N. D. Increased right amygdala activity during the hyper-acute phase of takotsubo syndrome. *Eur. Heart J.* **44**, 2260 (2023).
70. Khan, H. et al. Structural and functional brain changes in acute Takotsubo syndrome. *JACC Heart Fail.* **11**, 307–317 (2023).
71. Milner, T. A., Lubbers, L. S., Alves, S. E. & McEwen, B. S. Nuclear and extranuclear estrogen binding sites in the rat forebrain and autonomic medullary areas. *Endocrinology* **149**, 3306–3312 (2008).
72. van Wingen, G., Mattern, C., Verkes, R. J., Buitelaar, J. & Fernández, G. Testosterone reduces amygdala–orbitofrontal cortex coupling. *Psychoneuroendocrinology* **35**, 105–113 (2010).
73. Goldstein, J. M. et al. Normal sexual dimorphism of the adult human brain assessed by in vivo magnetic resonance imaging. *Cereb. Cortex* **11**, 490–497 (2001).
74. Haider, A. et al. Age- and sex-dependent changes of resting amygdalar activity in individuals free of clinical cardiovascular disease. *J. Nucl. Cardiol.* **28**, 427–432 (2021).
75. Mezue, K. et al. Reduced stress-related neural network activity mediates the effect of alcohol on cardiovascular risk. *J. Am. Coll. Cardiol.* **81**, 2315–2325 (2023).
76. Dar, T. et al. Greater neurobiological resilience to chronic socioeconomic or environmental stressors associates with lower risk for cardiovascular disease events. *Circ. Cardiovasc. Imaging* **13**, e010337 (2020).
77. Kang, D. O. et al. Stress-associated neurobiological activity is linked with acute plaque instability via enhanced macrophage activity: a prospective serial 18F-FDG-PET/CT imaging assessment. *Eur. Heart J.* **42**, 1883–1895 (2021).
78. Tawakol, A. et al. Relation between resting amygdalar activity and cardiovascular events: a longitudinal and cohort study. *Lancet* **389**, 834–845 (2017).
79. Fiechter, M. et al. Sex-dependent association between inflammation, neural stress responses, and impaired myocardial function. *Eur. J. Nucl. Med. Mol. Imaging* **47**, 2010–2015 (2020).
80. Fiechter, M. et al. Association between resting amygdalar activity and abnormal cardiac function in women and men: a retrospective cohort study. *Eur. Heart J. Cardiovasc. Imaging* **20**, 625–632 (2019).
81. Mehta, P. K., Lima, B. B., Nelson, M. D. & Bairey Merz, C. N. Adverse cardiovascular outcomes in women: blame the amygdala? *Eur. Heart J. Cardiovasc. Imaging* **20**, 633–635 (2019).
82. Gebhard, C., Bengs, S., Haider, A. & Fiechter, M. The neuro-inflammatory-vascular circuit: evidence for a sex-dependent interrelation? *Front. Neurosci.* **14**, 614345 (2020).
83. Dai, N. et al. Sex differences in coronary inflammation and atherosclerosis phenotypes in response to imaging marker of stress-related neural activity. *Circ. Cardiovasc. Imaging* **17**, e016057 (2024).
84. Mikail, N. & Gebhard, C. L'équilibre périlleux: mental stress and inflammation in women. *Circ. Cardiovasc. Imaging* **17**, e016525 (2024).
85. Civieri, G. et al. Anxiety and depression associated with increased cardiovascular disease risk through accelerated development of risk factors. *JACC Adv.* **3**, 101208 (2024).
86. Gebhard, C. et al. Impact of age and sex on left ventricular function determined by coronary computed tomographic angiography: results from the prospective multicentre CONFIRM study. *Eur. Heart J. Cardiovasc. Imaging* **18**, 990–1000 (2017).
87. Gebhard, C. et al. Age- and gender-dependent left ventricular remodeling. *Echocardiography* **30**, 1143–1150 (2013).
88. Borlaug, B. A., Sharma, K., Shah, S. J. & Ho, J. E. Heart failure with preserved ejection fraction: JACC Scientific Statement. *J. Am. Coll. Cardiol.* **81**, 1810–1834 (2023).
89. Omerovic, E. et al. Pathophysiology of Takotsubo syndrome — a joint scientific statement from the Heart Failure Association Takotsubo Syndrome Study Group and Myocardial Function Working Group of the European Society of Cardiology — part 1: overview and the central role for catecholamines and sympathetic nervous system. *Eur. J. Heart Fail.* **24**, 257–273 (2022).
90. Wittstein, I. S. et al. Neurohumoral features of myocardial stunning due to sudden emotional stress. *N. Engl. J. Med.* **352**, 539–548 (2005).
91. Naegele, M. et al. Endothelial function and sympathetic nervous system activity in patients with Takotsubo syndrome. *Int. J. Cardiol.* **224**, 226–230 (2016).
92. Sykes, R., Ang, D. T. Y. & Berry, C. Takotsubo is an acute myocardial ischaemic syndrome. *Eur. Heart J.* **46**, 1582–1585 (2025).
93. Isogai, T., Matsui, H., Tanaka, H., Fushimi, K. & Yasunaga, H. Early β-blocker use and in-hospital mortality in patients with Takotsubo cardiomyopathy. *Heart* **102**, 1029–1035 (2016).
94. Raposeiras-Roubín, S. et al. Long-term prognostic impact of beta-blockers in patients with Takotsubo syndrome: results from the RETAKO Registry. *Rev. Port. Cardiol.* **42**, 237–246 (2023).
95. Rossello, X. et al. Beta-blockers after myocardial infarction: effects according to sex in the REBOOT trial. *Eur. Heart J.* https://doi.org/10.1093/eurheartj/ehaf673 (2025).
96. Brenner, R. et al. Clinical characteristics, sex hormones, and long-term follow-up in Swiss postmenopausal women presenting with Takotsubo cardiomyopathy. *Clin. Cardiol.* **35**, 340–347 (2012).
97. Rossi, A. et al. Heart–brain interactions in cardiac and brain diseases: why sex matters. *Eur. Heart J.* **43**, 3971–3980 (2022).
98. Ueyama, T. et al. Chronic estrogen supplementation following ovariectomy improves the emotional stress-induced cardiovascular responses by indirect action on the nervous system and by direct action on the heart. *Circ. J.* **71**, 565–573 (2007).
99. El-Battrawy, I. et al. Estradiol protection against toxic effects of catecholamine on electrical properties in human-induced pluripotent stem cell derived cardiomyocytes. *Int. J. Cardiol.* **254**, 195–202 (2018).
100. Sung, B. H., Ching, M., Izzo, J. L. Jr., Dandona, P. & Wilson, M. F. Estrogen improves abnormal norepinephrine-induced vasoconstriction in postmenopausal women. *J. Hypertens.* **17**, 523–528 (1999).
101. Heck, A. L. & Handa, R. J. Sex differences in the hypothalamic-pituitary-adrenal axis' response to stress: an important role for gonadal hormones. *Neuropsychopharmacology* **44**, 45–58 (2019).
102. Wenner, M. M. et al. Influence of age and estradiol on sympathetic nerve activity responses to exercise in women. *Med. Sci. Sports Exerc.* **54**, 408–416 (2022).
103. Patel, S. M., Chokka, R. G., Prasad, K. & Prasad, A. Distinctive clinical characteristics according to age and gender in apical ballooning syndrome (Takotsubo/stress cardiomyopathy): an analysis focusing on men and young women. *J. Card. Fail.* **19**, 306–310 (2013).
104. Salmoirago-Blotcher, E. et al. Reproductive history of women with Takotsubo cardiomyopathy. *Am. J. Cardiol.* **118**, 1922–1928 (2016).
105. Möller, C. et al. Comprehensive assessment of sex hormones in Takotsubo syndrome. *Int. J. Cardiol.* **250**, 11–15 (2018).
106. Scally, C. et al. Myocardial and systemic inflammation in acute stress-induced (Takotsubo) cardiomyopathy. *Circulation* **139**, 1581–1592 (2019).
107. Khan, H. et al. A systematic review of biomarkers in Takotsubo syndrome: a focus on better understanding the pathophysiology. *Int. J. Cardiol. Heart Vasc.* **34**, 100795 (2021).
108. Mirijello, A. et al. Takotsubo syndrome and inflammatory bowel diseases: does a link exist? *Dig. Dis.* **38**, 204–210 (2020).
109. Shah, R. M., Shah, M., Shah, S., Li, A. & Jauhar, S. Takotsubo syndrome and COVID-19: associations and implications. *Curr. Probl. Cardiol.* **46**, 100763 (2021).

110. Vallabhajosyula, S., Deshmukh, A. J., Kashani, K., Prasad, A. & Sakhuja, A. Tako-tsubo cardiomyopathy in severe sepsis: nationwide trends, predictors, and outcomes. *J. Am. Heart Assoc.* **7**, e009160 (2018).
111. Sato, T. et al. Takotsubo (ampulla-shaped) cardiomyopathy associated with microscopic polyangiitis. *Intern. Med.* **44**, 251–255 (2005).
112. Hinterdobler, J. et al. Acute mental stress drives vascular inflammation and promotes plaque destabilization in mouse atherosclerosis. *Eur. Heart J.* **42**, 4077–4088 (2021).
113. Khera, A. et al. Race and gender differences in C-reactive protein levels. *J. Am. Coll. Cardiol.* **46**, 464–469 (2005).
114. Sullivan, S. et al. Young women with coronary artery disease exhibit higher concentrations of interleukin-6 at baseline and in response to mental stress. *J. Am. Heart Assoc.* **7**, e010329 (2018).
115. Martin, E. A. et al. Sex differences in vascular and endothelial responses to acute mental stress. *Clin. Auton. Res.* **18**, 339–345 (2008).
116. Vaccarino, V. et al. Sex differences in mental stress-induced myocardial ischemia in young survivors of an acute myocardial infarction. *Psychosom. Med.* **76**, 171–180 (2014).
117. Poller, W. C. et al. Brain motor and fear circuits regulate leukocytes during acute stress. *Nature* **607**, 578–584 (2022).
118. Bruns, B. et al. Calcineurin signaling promotes takotsubo syndrome. *Nat. Cardiovasc. Res.* **2**, 645–655 (2023).
119. Gudenkauf, B., Goetsch, M. R., Vakil, R. M., Cingolani, O. & Adamo, L. Case report: steroid-responsive Takotsubo cardiomyopathy associated with cytokine storm and obstructive shock. *Front. Cardiovasc. Med.* **9**, 931070 (2022).
120. Srichawla, B. S. Dimethyl fumarate-induced Takotsubo cardiomyopathy in a patient with relapsing–remitting multiple sclerosis. *Cureus* **14**, e23789 (2022).
121. Sacchetti, D. C., Furie, K. L. & Yaghi, S. Cardioembolic stroke: mechanisms and therapeutics. *Semin. Neurol.* **37**, 326–338 (2017).
122. Scheitz, J. F. et al. Stroke-heart syndrome: recent advances and challenges. *J. Am. Heart Assoc.* **11**, e026528 (2022).
123. Blaszczyk, E. et al. Myocardial injury in patients with acute ischemic stroke detected by cardiovascular magnetic resonance imaging. *Eur. J. Radiol.* **165**, 110908 (2023).
124. Prosser, J. et al. Predictors of early cardiac morbidity and mortality after ischemic stroke. *Stroke* **38**, 2295–2302 (2007).
125. Ishiguchi, H. et al. Mortality risk in patients with cardiac complications following ischemic stroke: a report from the Virtual International Stroke Trials Archive. *J. Am. Heart Assoc.* **13**, e036799 (2024).
126. Stengl, H. et al. Frequency, associated variables, and outcomes of acute myocardial injury according to the fourth Universal Definition of Myocardial Infarction in patients with acute ischemic stroke. *Eur. Stroke J.* **7**, 413–420 (2022).
127. Scheitz, J. F. et al. Prognostic relevance of cardiac troponin T levels and their dynamic changes measured with a high-sensitivity assay in acute ischaemic stroke: analyses from the TRELAS cohort. *Int. J. Cardiol.* **177**, 886–893 (2014).
128. Nelde, A. et al. Data lake-driven analytics identify nocturnal non-dipping of heart rate as predictor of unfavorable stroke outcome at discharge. *J. Neurol.* **270**, 3810–3820 (2023).
129. Buckley, B. J. R. et al. Stroke-heart syndrome: incidence and clinical outcomes of cardiac complications following stroke. *Stroke* **53**, 1759–1763 (2022).
130. Hellwig, S. et al. Cardiac troponin and recurrent major vascular events after minor stroke or TIA. *Ann. Neurol.* **90**, 901–912 (2021).
131. Broersen, L. H. A. et al. High-sensitivity cardiac troponin T and cognitive function in patients with ischemic stroke. *Stroke* **51**, 1604–1607 (2020).
132. Ishiguchi, H. et al. Incidence and outcomes of patients with early cardiac complications after intracerebral hemorrhage: a report from VISTA. *Stroke* **55**, 2669–2676 (2024).
133. Hoad, K. L. et al. Stroke–heart syndrome: incidence and clinical outcomes of cardiac complications following intracerebral haemorrhage. *Eur. Stroke J.* **10**, 100–107 (2025).
134. Klammer, M. G. et al. Association between stroke lesion size and atrial fibrillation detected after stroke: an observational cohort study. *J. Am. Heart Assoc.* **13**, e035285 (2024).
135. Krause, T. et al. Stroke in right dorsal anterior insular cortex Is related to myocardial injury. *Ann. Neurol.* **81**, 502–511 (2017).
136. Winder, K. et al. Acute right insular ischaemic lesions and poststroke left ventricular dysfunction. *Stroke Vasc. Neurol.* **8**, 301–306 (2023).
137. Seifert, F. et al. Neuroanatomical correlates of severe cardiac arrhythmias in acute ischemic stroke. *J. Neurol.* **262**, 1182–1190 (2015).
138. Colivicchi, F., Bassi, A., Santini, M. & Caltagirone, C. Prognostic implications of right-sided insular damage, cardiac autonomic derangement, and arrhythmias after acute ischemic stroke. *Stroke* **36**, 1710–1715 (2005).
139. Craig, A. D. How do you feel — now? The anterior insula and human awareness. *Nat. Rev. Neurosci.* **10**, 59–70 (2009).
140. Beissner, F., Meissner, K., Bar, K. J. & Napadow, V. The autonomic brain: an activation likelihood estimation meta-analysis for central processing of autonomic function. *J. Neurosci.* **33**, 10503–10511 (2013).
141. Sposato, L. A. et al. First-ever ischemic stroke and increased risk of incident heart disease in older adults. *Neurology* **94**, e1559–e1570 (2020).
142. Simats, A. et al. Innate immune memory after brain injury drives inflammatory cardiac dysfunction. *Cell* **187**, 4637–4655.e26 (2024).
143. Veltkamp, R. et al. Experimental ischaemic stroke induces transient cardiac atrophy and dysfunction. *J. Cachexia Sarcopenia Muscle* **10**, 54–62 (2019).
144. Yan, T. et al. Inflammatory responses mediate brain–heart interaction after ischemic stroke in adult mice. *J. Cereb. Blood Flow. Metab.* **40**, 1213–1229 (2020).
145. Lin, H. B. et al. Macrophage-NLRP3 inflammasome activation exacerbates cardiac dysfunction after ischemic stroke in a mouse model of diabetes. *Neurosci. Bull.* **36**, 1035–1045 (2020).
146. Stengl, H. et al. How the brain impacts the heart: lessons from ischaemic stroke and other neurological disorders. *Heart* **111**, 99–108 (2025).
147. Korpelainen, J. T., Sotaniemi, K. A., Huikuri, H. V. & Myllya, V. V. Abnormal heart rate variability as a manifestation of autonomic dysfunction in hemispheric brain infarction. *Stroke* **27**, 2059–2063 (1996).
148. Stengl, H. et al. Heart rate turbulence in acute ischemic stroke. *Eur. Stroke J.* **9**, 226–234 (2024).
149. Sykora, M., Diedler, J., Rupp, A., Turcani, P. & Steiner, T. Impaired baroreceptor reflex sensitivity in acute stroke is associated with insular involvement, but not with carotid atherosclerosis. *Stroke* **40**, 737–742 (2009).
150. Bieber, M. et al. Stroke-induced chronic systolic dysfunction driven by sympathetic overactivity. *Ann. Neurol.* **82**, 729–743 (2017).
151. Meloux, A. et al. Ischemic stroke increases heart vulnerability to ischemia-reperfusion and alters myocardial cardioprotective pathways. *Stroke* **49**, 2752–2760 (2018).
152. Lin, H. B. et al. Monocyte-derived macrophages aggravate cardiac dysfunction after ischemic stroke in mice. *J. Am. Heart Assoc.* **13**, e034731 (2024).
153. Vornholz, L. et al. Acute heart failure after reperfused ischemic stroke: association with systemic and cardiac inflammatory responses. *Front. Physiol.* **12**, 782760 (2021).
154. Simats, A. & Liesz, A. Systemic inflammation after stroke: implications for post-stroke comorbidities. *EMBO Mol. Med.* **14**, e16269 (2022).
155. Cao, J. et al. DNA-sensing inflammasomes cause recurrent atherosclerotic stroke. *Nature* **633**, 433–441 (2024).
156. Stark, K. & Massberg, S. Interplay between inflammation and thrombosis in cardiovascular pathology. *Nat. Rev. Cardiol.* **18**, 666–682 (2021).
157. Jiang, M. et al. A temperature-ultrasound sensitive nanoparticle delivery system for exploring central neuroinflammation mechanism in stroke–heart syndrome. *J. Nanobiotechnology* **22**, 681 (2024).
158. Kang, Y. M., Zhang, Z. H., Xue, B., Weiss, R. M. & Felder, R. B. Inhibition of brain proinflammatory cytokine synthesis reduces hypothalamic excitation in rats with ischemia-induced heart failure. *Am. J. Physiol. Heart Circ. Physiol* **295**, H227–H236 (2008).
159. Chan, K. L., Poller, W. C., Swirski, F. K. & Russo, S. J. Central regulation of stress-evoked peripheral immune responses. *Nat. Rev. Neurosci.* **24**, 591–604 (2023).
160. Courties, G. et al. Ischemic stroke activates hematopoietic bone marrow stem cells. *Circ. Res.* **116**, 407–417 (2015).
161. Durgan, D. J., Lee, J., McCullough, L. D. & Bryan, R. M. Jr. Examining the role of the microbiota–gut–brain axis in stroke. *Stroke* **50**, 2270–2277 (2019).
162. Inderhees, J. & Schwaninger, M. Liver metabolism in ischemic stroke. *Neuroscience* **550**, 62–68 (2024).
163. Stevens, R. D. & Puybasset, L. The brain–lung–brain axis. *Intensive Care Med.* **37**, 1054–1056 (2011).
164. DeFilippis, A. P. et al. Assessment and treatment of patients with type 2 myocardial infarction and acute nonischemic myocardial injury. *Circulation* **140**, 1661–1678 (2019).
165. Chen, J. et al. MiR-126 affects brain–heart interaction after cerebral ischemic stroke. *Transl. Stroke Res.* **8**, 374–385 (2017).
166. Hermanns, N. et al. Molecular imaging of the brain–heart axis provides insights into cardiac dysfunction after cerebral ischemia. *Basic. Res. Cardiol.* **117**, 52 (2022).
167. Battaglini, D. et al. Brain–heart interaction after acute ischemic stroke. *Crit. Care* **24**, 163 (2020).
168. Buckley, B. J. R., Harrison, S. L., Lane, D. A., Hill, A. & Lip, G. Y. H. Stroke–heart syndrome: mechanisms, risk factors, and adverse cardiovascular events. *Eur. J. Prev. Cardiol.* **31**, e23–e26 (2024).
169. Rosso, M. et al. Sex differences in outcomes of acute myocardial injury after stroke. *J. Am. Heart Assoc.* **13**, e032755 (2024).
170. Sposato, L. A., Lam, M., Allen, B., Shariff, S. Z. & Saposnik, G. First-ever ischemic stroke and incident major adverse cardiovascular events in 93 627 older women and men. *Stroke* **51**, 387–394 (2020).
171. Rohner, M. M., Gebhard, C., Luft, A., Hänsel, M. & Wegener, S. Cardiac findings following cerebrovascular disease. *J. Am. Heart Assoc.* **13**, e034131 (2024).
172. Akyea, R. K. et al. Sex, age, and socioeconomic differences in nonfatal stroke incidence and subsequent major adverse outcomes. *Stroke* **52**, 396–405 (2021).
173. Nolte, C. H. et al. Type 1 myocardial infarction in patients with acute ischemic stroke. *JAMA Neurol.* **81**, 703–711 (2024).
174. Johansen, M. C. et al. Association between acute myocardial infarction and cognition. *JAMA Neurol.* **80**, 723–731 (2023).
175. Hammond, C. A. et al. Long-term cognitive decline after newly diagnosed heart failure: longitudinal analysis in the CHS (Cardiovascular Health Study). *Circ. Heart Fail.* **11**, e004476 (2018).
176. Cannon, J. A. et al. Cognitive impairment and heart failure: systematic review and meta-analysis. *J. Card. Fail.* **23**, 464–475 (2017).
177. Bonnesen, M. P. et al. Atrial fibrillation burden and cognitive decline in elderly patients undergoing continuous monitoring. *Am. Heart J.* **242**, 15–23 (2021).
178. Petersen, M. et al. Mapping the interplay of atrial fibrillation, brain structure, and cognitive dysfunction. *Alzheimers Dement.* **20**, 4512–4526 (2024).
179. Shore, S. et al. Trajectory of cognitive function after incident heart failure. *Circ. Heart Fail.* **18**, e011837 (2025).

180. Liang, J. et al. Association between onset age of coronary heart disease and incident dementia: a prospective cohort study. *J. Am. Heart Assoc.* **12**, e031407 (2023).
181. Singh-Manoux, A. et al. Atrial fibrillation as a risk factor for cognitive decline and dementia. *Eur. Heart J.* **38**, 2612–2618 (2017).
182. Mejia-Renteria, H. et al. Coronary microvascular dysfunction is associated with impaired cognitive function: the Cerebral-Coronary Connection study (C3 study). *Eur. Heart J.* **44**, 113–125 (2023).
183. Vrinceanu, T. et al. Pattern of atrial fibrillation and cognitive function in young patients with atrial fibrillation and low CHADS(2) score: insights from the BRAIN-AF trial. *Circ. Arrhythm. Electrophysiol.* **15**, e010462 (2022).
184. Huijts, M. et al. Cognitive impairment in heart failure: results from the Trial of Intensified versus standard Medical therapy in Elderly patients with Congestive Heart Failure (TIME-CHF) randomized trial. *Eur. J. Heart Fail.* **15**, 699–707 (2013).
185. Johansen, M. C. et al. Risk of dementia associated with atrial cardiopathy: the ARIC study. *J. Am. Heart Assoc.* **11**, e025646 (2022).
186. Kresge, H. A. et al. Subclinical compromise in cardiac strain relates to lower cognitive performances in older adults. *J. Am. Heart Assoc.* **7**, e007562 (2018).
187. Schneider, A. L. et al. High-sensitivity cardiac troponin T and cognitive function and dementia risk: the atherosclerosis risk in communities study. *Eur. Heart J.* **35**, 1817–1824 (2014).
188. von Rennenberg, R. et al. High-sensitivity cardiac troponin T and cognitive function over 12 months after stroke-results of the DEMDAS study. *J. Am. Heart Assoc.* **13**, e033439 (2024).
189. Jensen, M. et al. Association of cardiac biomarkers with structural brain changes and cognitive impairment: results from the Hamburg City Health Study. *Neurology* **105**, e213865 (2025).
190. [No authors listed] Cardiogenic dementia. *Lancet* **1**, 27–28 (1977).
191. Vermeer, S. E. et al. Silent brain infarcts and the risk of dementia and cognitive decline. *N. Engl. J. Med.* **348**, 1215–1222 (2003).
192. Iadecola, C. et al. Vascular cognitive impairment and dementia: JACC Scientific Expert Panel. *J. Am. Coll. Cardiol.* **73**, 3326–3344 (2019).
193. Conen, D. et al. Relationships of overt and silent brain lesions with cognitive function in patients with atrial fibrillation. *J. Am. Coll. Cardiol.* **73**, 989–999 (2019).
194. Zhou, Z. et al. Covert cerebrovascular changes in people with heart disease: a systematic review and meta-analysis. *Neurology* **102**, e209204 (2024).
195. Mueller, K. et al. Brain damage with heart failure: cardiac biomarker alterations and gray matter decline. *Circ. Res.* **126**, 750–764 (2020).
196. Kuhne, M. et al. Silent brain infarcts impact on cognitive function in atrial fibrillation. *Eur. Heart J.* **43**, 2127–2135 (2022).
197. Stegmann, T. et al. Atrial fibrillation and atherosclerosis cause different vascular brain lesions on magnetic resonance imaging. *Eur. Heart J.* **46**, 5177–5188 (2025).
198. Alber, J. et al. White matter hyperintensities in vascular contributions to cognitive impairment and dementia (VCID): knowledge gaps and opportunities. *Alzheimers Dement. (N. Y.)* **5**, 107–117 (2019).
199. Voorter, P. H. M. et al. Heterogeneity and penumbra of white matter hyperintensities in small vessel diseases determined by quantitative MRI. *Stroke* **56**, 128–137 (2025).
200. Wardlaw, J. M., Smith, C. & Dichgans, M. Small vessel disease: mechanisms and clinical implications. *Lancet Neurol.* **18**, 684–696 (2019).
201. Keller, J. A. et al. Different cardiovascular risk factors are related to distinct white matter hyperintensity MRI phenotypes in older adults. *Neuroimage Clin.* **35**, 103131 (2022).
202. Stegmann, T. et al. Heart failure is independently associated with white matter lesions: insights from the population-based LIFE-Adult Study. *ESC Heart Fail.* **8**, 697–704 (2021).
203. Chen, L. Y. et al. Association of atrial fibrillation with cognitive decline and dementia over 20 years: the ARIC-NCS (Atherosclerosis Risk in Communities Neurocognitive Study). *J. Am. Heart Assoc.* **7**, e007301 (2018).
204. Frey, A. et al. Cognitive deficits and related brain lesions in patients with chronic heart failure. *JACC Heart Fail.* **6**, 583–592 (2018).
205. Frey, A. et al. Temporal changes in total and hippocampal brain volume and cognitive function in patients with chronic heart failure-the COGNITION.MATTERS-HF cohort study. *Eur. Heart J.* **42**, 1569–1578 (2021).
206. Mueller, K. et al. Brain dysconnectivity with heart failure. *Brain Commun.* **5**, fcad103 (2023).
207. Eichenbaum, H. On the integration of space, time, and memory. *Neuron* **95**, 1007–1018 (2017).
208. Klaassens, B. L. et al. Diminished posterior precuneus connectivity with the default mode network differentiates normal aging from Alzheimer's disease. *Front. Aging Neurosci.* **9**, 97 (2017).
209. Terasawa, Y., Fukushima, H. & Umeda, S. How does interoceptive awareness interact with the subjective experience of emotion? An fMRI study. *Hum. Brain Mapp.* **34**, 598–612 (2013).
210. Candia-Rivera, D. Brain–heart interactions in the neurobiology of consciousness. *Curr. Res. Neurobiol.* **3**, 100050 (2022).
211. Santamaria-Garcia, H. et al. Allostatic interoceptive overload across psychiatric and neurological conditions. *Biol. Psychiatry* **97**, 28–40 (2025).
212. Berntson, G. G. & Khalsa, S. S. Neural circuits of interoception. *Trends Neurosci.* **44**, 17–28 (2021).
213. Giannoni, A. et al. Chemoreflex and baroreflex sensitivity hold a strong prognostic value in chronic heart failure. *JACC Heart Fail.* **10**, 662–676 (2022).
214. Salah, H. M. et al. Baroreflex function in cardiovascular disease. *J. Card. Fail.* **31**, 117–126 (2025).
215. Goldberger, J. J., Arora, R., Buckley, U. & Shivkumar, K. Autonomic nervous system dysfunction: JACC Focus Seminar. *J. Am. Coll. Cardiol.* **73**, 1189–1206 (2019).
216. Ma, Y. et al. Baroreflex sensitivity and long-term dementia risk in older adults. *Hypertension* **82**, 347–356 (2025).
217. Schaich, C. L. et al. Association of heart rate variability with cognitive performance: the Multi-Ethnic Study of Atherosclerosis. *J. Am. Heart Assoc.* **9**, e013827 (2020).
218. Frysinger, R. C. & Harper, R. M. Cardiac and respiratory correlations with unit discharge in human amygdala and hippocampus. *Electroencephalogr. Clin. Neurophysiol.* **72**, 463–470 (1989).
219. Kim, D. et al. Less dementia after catheter ablation for atrial fibrillation: a nationwide cohort study. *Eur. Heart J.* **41**, 4483–4493 (2020).
220. Lin, F. V. & Heffner, K. L. Autonomic nervous system flexibility for understanding brain aging. *Ageing Res. Rev.* **90**, 102016 (2023).
221. McCracken, C. et al. Multi-organ imaging demonstrates the heart–brain–liver axis in UK Biobank participants. *Nat. Commun.* **13**, 7839 (2022).
222. Raisi-Estabragh, Z. et al. Associations of cognitive performance with cardiovascular magnetic resonance phenotypes in the UK Biobank. *Eur. Heart J. Cardiovasc. Imaging* **23**, 663–672 (2022).
223. Rajeev, V. et al. Chronic cerebral hypoperfusion: a critical feature in unravelling the etiology of vascular cognitive impairment. *Acta Neuropathol. Commun.* **11**, 93 (2023).
224. Endisch, C. et al. Hypoxic-ischemic encephalopathy evaluated by brain autopsy and neuroprognostication after cardiac arrest. *JAMA Neurol.* **77**, 1430–1439 (2020).
225. Gardarsdottir, M. et al. Atrial fibrillation is associated with decreased total cerebral blood flow and brain perfusion. *Europace* **20**, 1252–1258 (2018).
226. Gruhn, N. et al. Cerebral blood flow in patients with chronic heart failure before and after heart transplantation. *Stroke* **32**, 2530–2533 (2001).
227. Bhat, G., Yost, G. & Mahoney, E. Cognitive function and left ventricular assist device implantation. *J. Heart Lung Transpl.* **34**, 1398–1405 (2015).
228. Murphy, S. P., Kakkar, R., McCarthy, C. P. & Januzzi, J. L. Jr. Inflammation in heart failure: JACC state-of-the-art review. *J. Am. Coll. Cardiol.* **75**, 1324–1340 (2020).
229. Hu, Y. F., Chen, Y. J., Lin, Y. J. & Chen, S. A. Inflammation and the pathogenesis of atrial fibrillation. *Nat. Rev. Cardiol.* **12**, 230–243 (2015).
230. Kipinoinen, T. et al. Association of midlife inflammatory markers with cognitive performance at 10-year follow-up. *Neurology* **99**, e2294–e2302 (2022).
231. Cunningham, C. & Hennessy, E. Co-morbidity and systemic inflammation as drivers of cognitive decline: new experimental models adopting a broader paradigm in dementia research. *Alzheimers Res. Ther.* **7**, 33 (2015).
232. Koren, T. & Rolls, A. Immunoception: defining brain-regulated immunity. *Neuron* **110**, 3425–3428 (2022).
233. Li, Q. et al. Cardiac injury regulates neuroinflammation through extracellular vesicle-mediated heart–brain crosstalk. *JACC Basic. Transl. Sci.* **10**, 101307 (2025).
234. Wang, X. et al. Exosomes activate hippocampal microglia in atrial fibrillation through long-distance heart–brain communication. *BMC Cardiovasc. Disord.* **24**, 627 (2024).
235. Valenza, G. Depression as a cardiovascular disorder: central-autonomic network, brain–heart axis, and vagal perspectives of low mood. *Front. Netw. Physiol.* **3**, 1125495 (2023).
236. Celano, C. M., Daunis, D. J., Lokko, H. N., Campbell, K. A. & Huffman, J. C. Anxiety disorders and cardiovascular disease. *Curr. Psychiatry Rep.* **18**, 101 (2016).
237. Volgman, A. S. et al. Sex differences in cardiovascular disease and cognitive impairment: another health disparity for women? *J. Am. Heart Assoc.* **8**, e013154 (2019).
238. Dong, C. et al. Sex differences in the association between cardiovascular diseases and dementia subtypes: a prospective analysis of 464,616 UK Biobank participants. *Biol. Sex. Differ.* **13**, 21 (2022).
239. Delco, A. et al. Impact of sex and gender on heart failure. *Cardiovasc. Med.* **26**, 88–94 (2023).
240. Zilberman, J. M. et al. Association between hypertension, menopause, and cognition in women. *J. Clin. Hypertens.* **17**, 970–976 (2015).
241. Iadecola, C. et al. Impact of hypertension on cognitive function: a scientific statement from the American Heart Association. *Hypertension* **68**, e67–e94 (2016).
242. Doehner, W. et al. Heart and brain interaction in patients with heart failure: overview and proposal for a taxonomy. A position paper from the Study Group on Heart and Brain Interaction of the Heart Failure Association. *Eur. J. Heart Fail.* **20**, 199–215 (2018).
243. Roy, B. et al. Reduced regional cerebral blood flow in patients with heart failure. *Eur. J. Heart Fail.* **19**, 1294–1302 (2017).
244. Jacobs, E. G. Only 0.5% of neuroscience studies look at women's health. Here's how to change that. *Nature* **623**, 667 (2023).
245. Buckley, R. F., Gong, J. & Woodward, M. A call to action to address sex differences in Alzheimer disease clinical trials. *JAMA Neurol.* **80**, 769–770 (2023).
246. Hermann, B. et al. Aberrant brain–heart coupling is associated with the severity of post cardiac arrest brain injury. *Ann. Clin. Transl. Neurol.* **11**, 866–882 (2024).
247. Keynan, J. N. et al. Electrical fingerprint of the amygdala guides neurofeedback training for stress resilience. *Nat. Hum. Behav.* **3**, 63–73 (2019).
248. Van Gelder, I. C. et al. 2024 ESC guidelines for the management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS). *Eur. Heart J.* **45**, 3314–3414 (2024).
249. Lip, G. Y. H. et al. Integrated care for optimizing the management of stroke and associated heart disease: a position paper of the European Society of Cardiology Council on Stroke. *Eur. Heart J.* **43**, 2442–2460 (2022).

250. Mohanta, S. K. et al. Neuroimmune cardiovascular interfaces control atherosclerosis. *Nature* **605**, 152–159 (2022).
251. Yaqub, A. et al. Clinical and imaging markers of cardiac function and brain health: a meta-analysis of community-based studies. *Neurology* **104**, e213421 (2025).
252. Smith, J. R. et al. Contribution of modifiable midlife and late-life vascular risk factors to incident dementia. *JAMA Neurol.* **82**, 644–654 (2025).

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